VISUAL SENSITIVITY IS NOT CLAIM RETRACTABILITY: PERSISTENCE-AWARE CREDIT ASSIGNMENT FOR MULTIMODAL REINFORCEMENT LEARNING

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ABSTRACT

Reinforcement Learning with Verifiable Rewards (RLVR) has been extended to Large Vision–Language Models (LVLMs), and perception-aware methods further encourage policies to rely on visual evidence. Yet relying on the image does not guarantee that visual claims are supported by it. Before RL training, 27.81% of the correctly answered responses of Qwen2.5-VL-7B on four multimodal reasoning benchmarks contain at least one direct visual claim that the image does not support. Since outcome-level RL rewards each response as a whole, these claims inherit the positive credit of the correct answer. We introduce a fixed-rollout counterfactual diagnostic that re-scores the same response under an intervened image to separate Evidence-Function Sensitivity (EFS), how strongly the model's predictions change, from claim persistence, whether the model keeps supporting the same claim rather than retracting it. The diagnostic reveals Sensitivity–Persistence Decoupling (SPD): under DAPO and VPPO, EFS increases and claims become more retractable overall, yet unsupported claims become significantly more persistent, whereas GRPO raises EFS without this deterioration. We therefore propose Persistence-Aware Credit Gating (PACG), which attenuates positive credit for unusually persistent visual claims and leaves all other credit unchanged. It requires no supported/unsupported labels and adds no inference cost. On Qwen2.5-VL-7B, PACG raises the nine-benchmark average over three seeds from 58.1% to 59.9% with DAPO and from 59.8% to 60.9% with VPPO, while making unsupported claims more retractable. The gains extend to a larger model, a newer backbone, and the accuracy of HallusionBench also improves consistently. These results suggest that visual sensitivity and claim retractability are complementary dimensions of multimodal credit assignment.

1 INTRODUCTION

Reinforcement Learning with Verifiable Rewards (RLVR) has driven substantial progress in language model post-training (Shao et al., 2024; Guo et al., 2025; Yu et al., 2026a; Jaech et al., 2024) and has been extended to Large Vision–Language Models (LVLMs) for visual mathematics, chart understanding, and geometric reasoning (Huang et al., 2025; Liu et al., 2025c; Meng et al., 2025; Zhang et al., 2025). Yet stronger reasoning performance does not imply reliable visual grounding. LVLMs can retain systematic visual blind spots (Tong et al., 2024), extended reasoning can amplify visual hallucination (Liu et al., 2025a), and RL gains may persist even under hallucination-inductive visual conditions (Zhang et al., 2026a). A single trajectory-level reward cannot identify which parts of a multimodal reasoning trace deserve reinforcement. To address this credit assignment problem, recent work redirects supervision toward visual evidence via perception rewards, implicit objectives, token-level reweighting, and visually grounded selection (Xiao et al., 2026; Wang et al., 2026b; Huang et al., 2026c; Li et al., 2026; Lu et al., 2026; Huang et al., 2026b; Jiao et al., 2026). Multimodal process reward models further provide finer-grained supervision (Wang et al., 2026a; Zhou et al., 2026). These approaches mainly determine where and how strongly a policy relies on the image, leaving open whether individual visual claims are actually supported by the visual evidence.

We study this question at the level of direct visual claims, the minimal response spans that assert information read directly from the image, such as counts, OCR values, or geometric properties.

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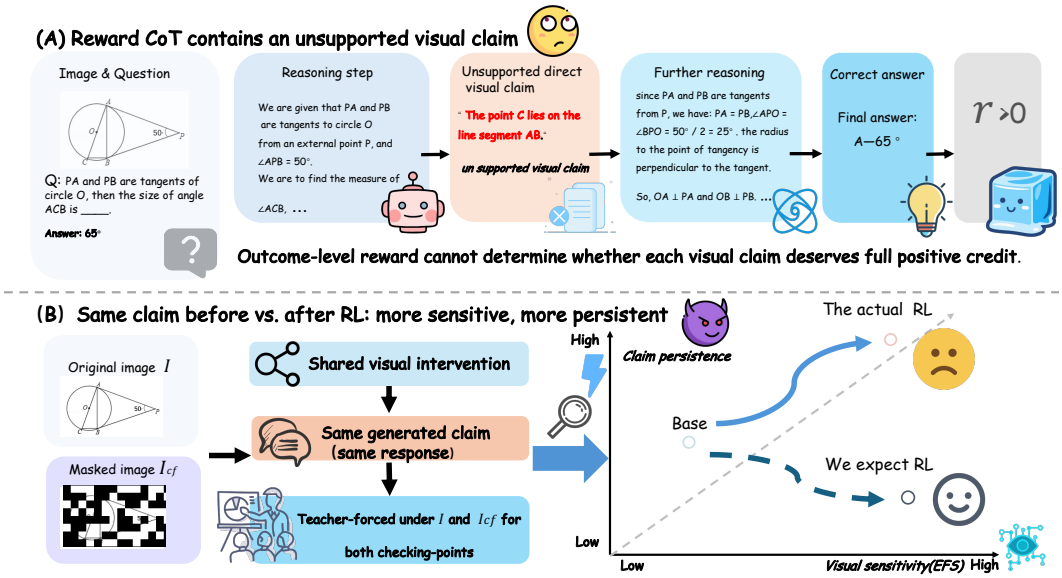


Figure 1: A correct pre-RL answer can conceal an unsupported visual assertion. We fix this rewarded trace and compare its teacher-forced behavior across checkpoints, asking whether outcome RL makes the claim more sensitive to image intervention without making it more retractable.

As shown in Figure 1(A), such a claim can appear inside a trajectory that still reaches the correct answer. We call a claim retractable if its support decreases when the visual evidence is disrupted, and persistent if it retains that support. The distinction matters because outcome-level RL rewards trajectories as a whole. We evaluate Qwen2.5-VL-7B on four multimodal reasoning benchmarks and collect the trajectories. We find that 2,370 of 8,521 answer-correct Base trajectories (27.81%) contain at least one unsupported direct visual claim, and each such claim inherits positive trajectory-level credit despite lacking visual support. One might expect that making a policy more visually sensitive would also make these claims easier to retract, but we find that it does not:

Greater visual sensitivity does not imply that unsupported visual claims become more retractable.

To address this, we introduce a fixed-rollout counterfactual diagnostic. For the same generated claim, we teacher-force the fixed response under the original image I and an intervened image I^{cf}. **Evidence-Function Sensitivity (EFS)** measures how strongly the predictive distribution at the claim tokens changes under the intervention, whereas **claim persistence** measures whether support for the same claim is retained after its visual evidence is disrupted. The latter is calibrated against residual tokens from the same rollout to remove trajectory-wide likelihood drift. Tracking both quantities across checkpoints reveals **Sensitivity–Persistence Decoupling (SPD)**. Under DAPO and VPPO, EFS increases and the overall retractability of visual claims improves, yet unsupported claims become more persistent. GRPO increases EFS without the same deterioration, which shows that the relation between sensitivity and retractability depends on the optimizer rather than being universal.

Building on this diagnosis, we propose **Persistence-Aware Credit Gating (PACG)**, which uses claim persistence to gate positive token credit. Routed direct visual claims that retract by a sufficient margin keep their full positive advantage, whereas unusually persistent claims receive attenuated positive credit. Negative advantages and non-routed tokens are left unchanged. PACG uses no supported/unsupported labels during training and adds no inference-time cost. Across nine multimodal reasoning benchmarks, PACG improves DAPO on all nine benchmarks and further improves VPPO, achieving the highest aggregate among our matched 7B runs, with both gains significant across three training seeds. With both parent optimizers, it significantly reduces unsupported-claim persistence while preserving or improving EFS and final-answer accuracy, including on HallusionBench.

Our contributions are threefold:

- We identify a credit-assignment failure in multimodal RL: unsupported direct visual claims frequently occur inside answer-correct trajectories and can inherit positive trajectory-level credit.
- We introduce a fixed-rollout counterfactual diagnostic that separates visual sensitivity from claim persistence and uncover Sensitivity–Persistence Decoupling: multimodal RL can increase visual sensitivity and aggregate retractability while making unsupported visual claims more persistent.
- We propose PACG, a conservative and composable credit-gating mechanism that improves unsupported-claim retractability across DAPO and VPPO while preserving or improving visual sensitivity, benchmark accuracy, and HallusionBench performance.

2 Related Work

Multimodal Reinforcement Learning with Verifiable Rewards. RLVR improves language-model reasoning by optimizing policies against verifiable outcome rewards, with GRPO and its large-scale variants estimating advantages from group-relative rewards (Shao et al., 2024; Guo et al., 2025; Yu et al., 2026a). Recent work extends RLVR to vision–language models through task-specific rewards and multimodal cold starts (Liu et al., 2025c; Huang et al., 2025), multi-stage perception–reasoning training (Meng et al., 2025; Peng et al., 2025; Feng et al., 2026), and iterative SFT–RL optimization (Zhang et al., 2025; Deng et al., 2025). Most of these methods still broadcast a single answer-level advantage across all response tokens. We show that this lets unsupported visual claims inside correct trajectories inherit positive credit, and that the resulting persistence shift depends on the optimizer: DAPO and VPPO worsen it, whereas GRPO does not have this effect.

Perception-Aware and Fine-Grained Policy Optimization. To make credit more visually grounded, prior work introduces explicit perception rewards (Xiao et al., 2026), implicit perception objectives from visually suppressed inputs (Wang et al., 2026b), token-level reweighting or selection of visually dependent tokens (Huang et al., 2026c; Jiao et al., 2026), and counterfactual or role-aware objectives against language priors (Jiang & Zou, 2026; Yu et al., 2026b). Multimodal process reward models further supervise individual steps (Wang et al., 2026a; Zhang et al., 2026b), and grounding-oriented methods tie reasoning to localized evidence regions (Cao et al., 2025; Huang et al., 2026a). These methods control where and how strongly the policy relies on the image. PACG addresses a complementary question, namely whether a specific visual claim loses support when its evidence is disrupted, and it can be composed directly with perception-aware methods such as VPPO.

Multimodal Hallucination and Counterfactual Faithfulness. Visual hallucination is typically evaluated with object, attribute, and relation benchmarks (Li et al., 2023; Wang et al., 2023; Guan et al., 2024) and mitigated through preference optimization or fine-grained feedback (Yu et al., 2024; Zhou et al., 2024; Sarkar et al., 2025; Yu et al., 2025), or at inference time through contrastive decoding (Leng et al., 2024; Huang et al., 2024). Counterfactual interventions have also been used to analyze modality-specific causal effects, hallucination, and explanation faithfulness (Li et al., 2025; Ding et al., 2025; Zafar et al., 2026). These approaches either evaluate a fixed model or modify decoding and preference data. In contrast, we use counterfactual image intervention inside RL training, turning the retractability of each reasoning claim into a token-level credit signal, as this training-only mechanism adds no inference-time computational or latency cost.

3 Preliminaries

Group-relative RLVR. Given a multimodal prompt $x = (q, I)$, the rollout policy $\pi_{\theta_{\text{old}}}$ samples a group of responses $\{y_i\}_{i=1}^{G}$, and each response receives a verifiable reward $r_i \in \{0, 1\}$ for final-answer correctness. GRPO and DAPO turn these rewards into group-relative advantages (Shao et al., 2024; Yu et al., 2026a) and optimize a clipped objective, written here in DAPO's token-mean form:

$$\mathcal{J}(\theta) = \mathbb{E}\left[\frac{1}{\sum_i |y_i|}\sum_{i=1}^{G}\sum_{t=1}^{|y_i|}\min(\rho_{i,t}A_{i,t},\ \text{clip}(\rho_{i,t}, 1-\epsilon_{\text{low}}, 1+\epsilon_{\text{high}})A_{i,t})\right], \tag{1}$$

where $\rho_{i,t} = \pi_\theta(y_{i,t} \mid x, y_{i,<t})/\pi_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})$ denotes the importance ratio and $A_{i,t}$ is the credit assigned to token t. GRPO differs only in averaging the loss per sequence and in using symmetric clipping, and we omit parent-specific regularizers.

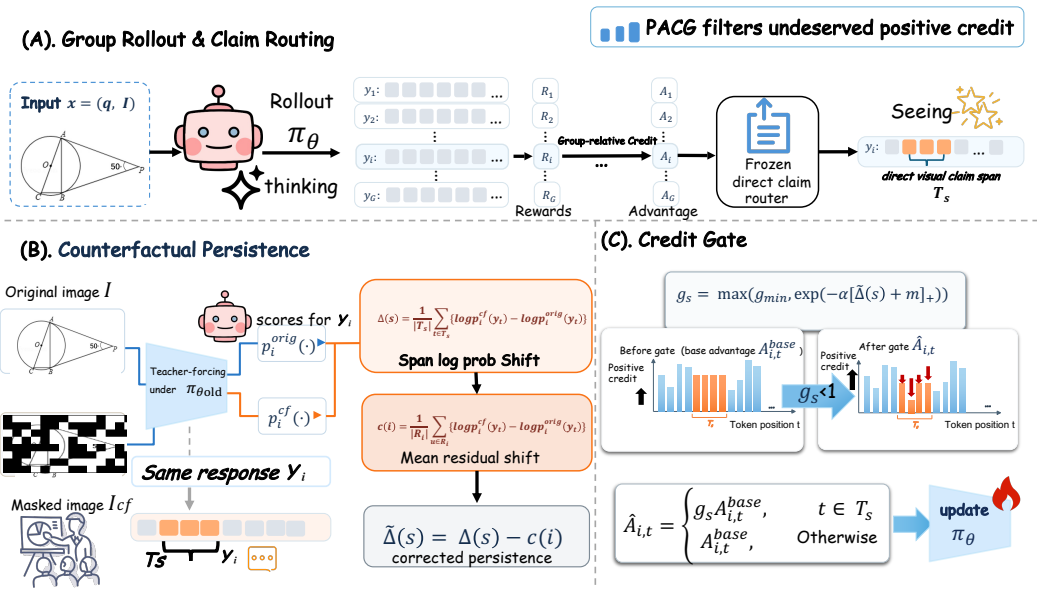


Figure 2: **Overview of PACG.** A frozen router identifies direct visual claims in responses scored under the original and degraded images. The residual-corrected signed shift $\tilde{\Delta}(s)$ measures claim persistence and determines a soft gate on positive token credit. Negative advantages and non-routed tokens are unchanged, so the method only modulates positive credit for routed visual claims.

Outcome-level credit ambiguity. What matters for this work is how the token credit $A_{i,t}$ is obtained. Outcome-driven optimizers such as GRPO and DAPO simply broadcast the sequence advantage to every token, assigning each token the same sequence-level advantage:

$$A^{\text{base}}_{i,t} = A_i, \tag{2}$$

while perception-aware optimizers such as VPPO reshape this credit with image-perturbation dependency scores (Huang et al., 2026c). In both cases, the credit a visual claim receives depends on whether the final answer is correct and on how strongly its tokens depend on the image, but not on whether the image actually supports the claim. An unsupported claim inside a correct trajectory therefore inherits positive credit, and the objective provides no signal to distinguish it from a supported one. PACG is designed to supply such a signal without requiring support labels.

4 Method

As illustrated in Figure 2, PACG proceeds in three steps. It first identifies the spans of a response that make direct visual claims, then measures whether each claim retracts when its visual evidence is disrupted, and finally scales down the positive credit of claims that persist. Only the positive parent credit on routed claims is modified during optimization, while all other credit remains unchanged.

4.1 Direct Visual Claim Routing

We focus on direct visual claims, the minimal spans of a response that assert information read directly from the image, such as an object attribute, a count, an OCR value, a spatial relation, a chart reading, or a geometric property. As shown in Figure 2(A), this definition is functional rather than factual. An incorrect assertion is still routed if it presents itself as image evidence, whereas derived reasoning and paraphrases of the question are excluded. A frozen Qwen3.6-35B-A3B router maps each claim s to its token set $\mathcal{T}_s$ in response y_i. After excluding routed and otherwise visually dependent tokens, the remaining tokens form a residual set $\mathcal{R}_i$ that serves as a within-response reference. Whether a claim is supported by the image is annotated only for analysis and never used in training.

4.2 FIXED-ROLLOUT COUNTERFACTUAL MEASUREMENT

To test whether a claim depends on its evidence, we degrade the image, $I_i^{\mathrm{cf}} = \mathcal{C}(I_i)$, and teacher-force the same sampled response under both images with a frozen policy π_ϕ, as shown in Figure 2(B). Because every response prefix is fixed, any change in the predictions of the model comes from the visual context alone. We write $p_{i,t}^{\mathrm{orig}}(\cdot) = \pi_\phi(\cdot \mid q_i, I_i, y_{i,<t})$ and $p_{i,t}^{\mathrm{cf}}(\cdot) = \pi_\phi(\cdot \mid q_i, I_i^{\mathrm{cf}}, y_{i,<t})$, where $\phi = \theta_{\mathrm{old}}$ during training and ϕ is the evaluated checkpoint in offline analysis.

Sensitivity. A natural first measure is how much the predictive distribution at the claim tokens changes. We define this change as Evidence-Function Sensitivity, or EFS for short.

$$\mathrm{EFS}_\phi(s) = \frac{1}{|\mathcal{T}_s|} \sum_{t \in \mathcal{T}_s} D_{\mathrm{KL}}\Big(p_{i,t}^{\mathrm{orig}} \,\|\, p_{i,t}^{\mathrm{cf}}\Big). \tag{3}$$

EFS, however, is unsigned. A large divergence can arise when the claim becomes less likely, but also when it becomes more likely, for instance when the masked image leads the model to fall back on a language prior that favors the same assertion. EFS therefore reveals how strongly a claim reacts to the image but not whether the model withdraws it, and we use it only for diagnosis.

Persistence. We track the claim's token log-likelihood to capture this directional change:

$$\Delta_\phi(s) = \frac{1}{|\mathcal{T}_s|} \sum_{t \in \mathcal{T}_s} \Big[\log p_{i,t}^{\mathrm{cf}}(y_{i,t}) - \log p_{i,t}^{\mathrm{orig}}(y_{i,t})\Big]. \tag{4}$$

Image corruption also shifts the likelihood of the entire response, so a negative $\Delta_\phi(s)$ may reflect trajectory-wide drift rather than a claim-specific retraction. We therefore correct for this drift by subtracting the mean shift of the residual tokens in the same response:

$$c_\phi(i) = \frac{1}{|\mathcal{R}_i|} \sum_{u \in \mathcal{R}_i} \Big[\log p_{i,u}^{\mathrm{cf}}(y_{i,u}) - \log p_{i,u}^{\mathrm{orig}}(y_{i,u})\Big], \qquad \widetilde{\Delta}_\phi(s) = \Delta_\phi(s) - c_\phi(i). \tag{5}$$

A negative $\widetilde{\Delta}_\phi(s)$ means that the claim loses more support than the rest of the response, which we call retraction, whereas values near or above zero indicate persistence. The residual set is a practical reference, and the intervention tests how a claim responds to its evidence. We refer to the mismatch between EFS and corrected persistence as Sensitivity–Persistence Decoupling (SPD).

4.3 PERSISTENCE-AWARE CREDIT GATING

As shown in Figure 2(C), the corrected persistence of each routed claim soft-gates:

$$v_s = [\widetilde{\Delta}_{\theta_{\mathrm{old}}}(s) + m]_+, \qquad g_s = \max\big(g_{\min}, e^{-\alpha v_s}\big), \tag{6}$$

where $m \geq 0$ is the retraction margin, $\alpha \geq 0$ controls the strength of attenuation, and $g_{\min} \in (0, 1]$ is the credit floor. Claims that retract by at least m keep their full positive credit, while more persistent claims are attenuated smoothly toward $g_{\min}$, so even the most persistent claim retains part of its credit. The gate is computed with the frozen rollout policy and treated as a stop-gradient, which prevents the policy from raising its credit by differentiating through the gate.

4.4 PACG POLICY OPTIMIZATION

Finally, the gates are applied to the token credit of the parent. Let $g_{i,t}^{\mathrm{span}}$ be the minimum gate among routed claims covering token t, with value one for non-routed tokens. PACG replaces the original unscaled parent credit with a soft-gated credit computed as follows:

$$\widetilde{A}_{i,t} = \begin{cases} g_{i,t}^{\mathrm{span}} A_{i,t}^{\mathrm{base}}, & A_{i,t}^{\mathrm{base}} > 0 \text{ and } t \text{ is routed}, \\ A_{i,t}^{\mathrm{base}}, & \text{otherwise}, \end{cases} \tag{7}$$

and substitutes $\widetilde{A}_{i,t}$ for $A_{i,t}$ in Equation 1, leaving the reward, rollout generation, normalization, and clipping unchanged. For GRPO and DAPO, $A_{i,t}^{\mathrm{base}}$ is the broadcast credit of Equation 2, and for VPPO it is the perception-shaped token credit.

Why positive-only gating? Negative advantages already suppress tokens from failed trajectories, including their persistent claims. Attenuating them would weaken this corrective signal without establishing that any individual claim is wrong, so PACG gates only positive credit. Because $g_{i,t}^{\text{span}} \in [g_{\min}, 1]$, PACG preserves the sign of every credit, never increases its magnitude, and reduces to the parent optimizer when no claim is routed or when all routed claims retract.

Can the policy evade the gate? A policy might try to avoid attenuation by producing fewer visual claims. A free-generation audit shows only a modest drop in the rate of direct claims together with longer responses, rather than a collapse of visual content. Routing and fallback rules, this audit, the formal properties of the gate, and the complete training procedure are provided in the appendix.

5 Experiments

5.1 Experimental Setup

Models and training. Following VPPO, we train Qwen2.5-VL-7B on ViRL39K (Wang et al., 2025a), comparing Base, GRPO, DAPO, VPPO, and PACG compositions. We also evaluate Qwen2.5-VL-32B and Qwen3-VL-2B-Thinking. Public baselines include ThinkLite-VL, VL-Rethinker, Noisy-Rollout, R1-ShareVL, and MM-Eureka (Wang et al., 2025b;a; Liu et al., 2025b; Yao et al., 2025; Meng et al., 2025). The 7B and 32B runs use two epochs, learning rate 10^{-6}, rollout batch size 384, and response limits of 2,048 and 4,096 tokens, while the 2B runs use 200 steps and 8,192 tokens. All runs use an entropy penalty of 0.1, and PACG uses $m = 0.05$, $\alpha = 10$, $g_{\min} = 0.30$, and random 14×14 patch blackening with probability 0.5. Further training details are provided in Appendix A.

Evaluation. Task-level accuracy is evaluated on MMK12 (Meng et al., 2025), MathVerse (Zhang et al., 2024), DynaMath (Zou et al., 2024), MathVision (Wang et al., 2024), Geometry3K (Lu et al., 2021), We-Math (Qiao et al., 2024), LogicVista (Xiao et al., 2024), MMMU-Pro (Yue et al., 2024), and Clever-Count (Johnson et al., 2017). We follow the evaluation protocol of VPPO (Huang et al., 2026c) and report mean accuracy over eight responses at temperature 1.0, with Avg. denoting the unweighted mean over the nine benchmarks. PACG adds no inference cost and increases the 7B training time per step by 57.3%, as detailed in Appendix H.

5.2 Main Results

Q1: Does persistence-aware gating improve task accuracy? Yes, for both parent optimizers. As shown in Table 1, DAPO+PACG improves all nine benchmarks over DAPO and raises the average from 58.1 to 59.9, while VPPO+PACG improves eight benchmarks and reaches the best 7B average of 60.9. The largest gains appear on Clever-Count and MathVerse, the benchmarks that depend most directly on reading the image. Because PACG leaves the reward untouched, these gains come from credit allocation alone. Both average gains are statistically significant across three training seeds under a two-sided Welch's t-test ($p < 0.05$), and the per-seed results are reported in Appendix F.3.

Q2: Is the effect specific to one model? No. As shown in Table 1, DAPO+PACG raises the average from 66.3 to 67.2 on Qwen2.5-VL-32B and from 51.3 to 53.5 on Qwen3-VL-2B-Thinking, where VPPO+PACG again gives the best aggregate (54.5).

Q3: Does the gain carry over to a hallucination benchmark? Yes, consistently though modestly. Table 2 shows that PACG improves all three HallusionBench (Guan et al., 2024) metrics for both parents, and VPPO+PACG is best on every column. The largest gain for VPPO is 0.86 points on qAcc, which requires the same question to be answered correctly across its visual variants. fAcc, which requires every question about an image to be correct, also rises for both parents, so the gains are not confined to isolated questions.

Table 2: **HallusionBench** (7B; %; single checkpoint per model).

Model	aAcc↑	fAcc↑	qAcc↑
Base	65.19	38.15	33.41
GRPO	66.90	44.35	35.10
DAPO	67.17	43.77	35.32
VPPO	67.70	45.06	35.18
DAPO+PACG	67.64	44.64	35.54
VPPO+PACG	**68.11**	**45.51**	**36.04**

Table 1: **Main results across multimodal reasoning benchmarks and backbone families.** All entries are accuracy (%). The first six datasets assess general mathematical and geometric reasoning, whereas the last three emphasize vision-dependent multimodal reasoning. Public 7B baselines are listed first, followed by our matched Qwen2.5-VL-7B runs. **Bold** and underlined values denote the best and second-best completed results within each backbone block. Additional blocks show transfer to a larger model scale and a newer backbone family.

Backbone	Method	General Mathematical & Geometric Reasoning						Vision-Dependent Reasoning			Avg.
		MMK12	Math Verse	Dyna Math	Math Vision	Geometry 3K	We-Math	Logic Vista	MMMU-Pro	Clever-Count	
Qwen2.5-VL 7B	ThinkLite-VL	62.5	63.8	62.0	31.5	35.8	66.4	40.0	27.6	77.8	51.9
	VL-Rethinker	69.3	68.8	65.7	29.5	40.7	68.5	45.8	39.7	82.0	56.7
	NoisyRollout*	50.0	67.8	62.1	22.1	46.9	71.0	45.3	34.5	85.7	53.9
	R1-ShareVL	70.9	68.2	63.9	27.5	41.2	69.9	45.4	35.1	81.2	55.9
	MM-Eureka	67.5	65.4	64.8	27.1	40.8	65.5	45.8	35.3	77.8	54.4
	Base	42.8	40.2	55.0	18.8	35.5	50.2	40.6	30.0	61.6	41.6
	GRPO	72.2	67.8	65.1	30.1	42.0	67.8	47.1	38.3	81.0	56.8
	DAPO	81.0	66.6	64.6	30.3	42.4	67.9	46.2	39.2	84.6	58.1
	VPPO	80.4	70.1	66.5	31.0	44.7	70.1	47.9	40.0	87.4	59.8
	DAPO+PACG (Ours)	81.2	**70.6**	65.9	30.6	44.3	70.1	47.1	**40.7**	88.7	59.9
	VPPO+PACG (Ours)	**81.5**	70.1	**67.0**	**32.1**	**45.7**	**71.5**	**50.4**	40.6	**89.5**	**60.9**
Transfer to a Larger Model Scale											
Qwen2.5-VL 32B	Base	56.9	67.2	69.9	36.9	50.0	68.2	55.6	45.8	78.4	58.8
	GRPO	79.5	74.1	65.0	41.0	53.4	**73.3**	58.6	**48.6**	83.5	64.1
	DAPO	86.5	75.2	**74.3**	41.8	53.8	72.1	59.1	47.9	86.2	66.3
	DAPO+PACG (Ours)	**87.1**	**75.9**	73.8	**42.7**	**54.1**	72.8	**61.1**	47.9	**89.0**	**67.2**
Transfer to a Newer Backbone Family											
Qwen3-VL 2B Thinking	Base	31.1	51.5	53.1	19.0	36.4	57.2	32.0	20.4	74.1	41.6
	GRPO	51.3	53.5	62.0	24.4	45.4	57.1	33.2	23.3	83.2	48.1
	DAPO	50.8	60.1	61.5	25.1	44.2	65.6	43.7	24.9	86.2	51.3
	VPPO	**51.7**	68.9	**62.3**	**26.7**	47.2	67.6	44.5	25.7	88.6	53.7
	DAPO+PACG (Ours)	51.6	67.5	61.8	25.8	46.9	67.9	44.6	26.0	89.0	53.5
	VPPO+PACG (Ours)	**51.7**	**69.4**	62.1	**26.7**	**48.9**	**68.2**	**45.0**	**26.1**	**92.8**	**54.5**

*NoisyRollout is trained using the training set of Geometry3K; its Geometry3K score is excluded from that column's ranking.

Q4: Does gating positive credit slow down outcome learning? No. As shown in Figure 3, PACG closely tracks the reward trajectory of GRPO and raises the late-stage reward of DAPO and VPPO. Online entropy, length, and gate statistics are reported in Appendix G.

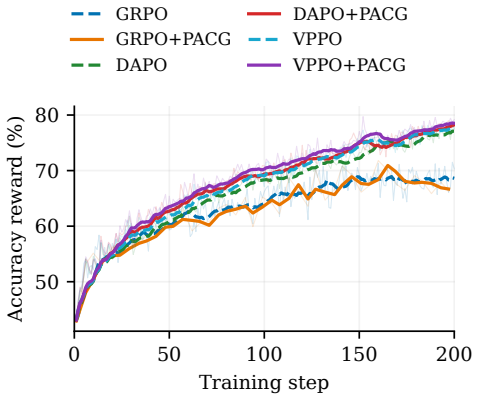


Figure 3: **Online accuracy reward.** Dashed: parents; solid: PACG compositions; curves are trailing 10-step means for one representative seed per method.

5.3 Ablation Study

Q5: Which components of the gate actually matter? All three. Table 3 shows that without direct-claim routing, gating every positive token destroys the training signal and accuracy collapses to 38.0. Removing residual calibration or the positive-only restriction remains competitive on the mathematical benchmarks but loses 7.4 and 6.1 points on Clever-Count and weakens unsupported-claim retractability, confirming that both components are necessary.

Table 3: **Component ablations with DAPO as parent.** *w/o Residual Corr.*: omit same-rollout residual calibration; *w/o Direct Routing*: gate all positive tokens instead of direct claims; *w/o Positive-Only*: also attenuate negative advantages. Accuracy and its nine-benchmark mean (Avg.) are percentages. Unsup. Pers. is the unsupported corrected persistence defined in Section 5.4. Entries are point estimates; bold marks the best column value.

Variant	MMK12	Math Verse	Dyna Math	Math Vision	Geometry 3K	We-Math	Logic Vista	MMMU-Pro	Clever-Count	Avg.	Unsup. Pers. ↓
DAPO	81.0	66.6	64.6	30.3	42.4	67.9	46.2	39.2	84.6	58.1	−0.1241
w/o Residual Corr.	80.9	69.0	65.6	30.9	**45.4**	69.1	**48.9**	40.1	81.3	59.0	−0.1278
w/o Direct Routing	46.3	40.8	36.7	16.2	10.4	44.0	31.7	30.3	85.2	38.0	−0.0374
w/o Positive-Only	80.8	70.3	**66.1**	**31.3**	44.5	**70.6**	47.7	39.6	82.6	59.3	−0.1301
PACG (full)	**81.2**	**70.6**	65.9	30.6	44.3	70.1	47.1	**40.7**	**88.7**	**59.9**	**−0.1392**

Margin selection. Table 4 compares three margins. The zero margin retains 95% of positive credit and matches the MMK12 accuracy of ungated DAPO. At $m = 0.10$, retention drops to 65% and EFS to 0.28, and accuracy falls slightly below the result at $m = 0.05$ despite stronger retraction. We therefore use $m = 0.05$ as the operating point.

Table 4: **Margin sweep** (α=10, $g_{\min}$=0.30). Acc.: MMK12; Ret.: positive-credit retention (both %).

m	Acc.	EFS ↑	Unsup. Pers. ↓	Ret.
0.00	81.0	**0.34**	−0.08	95
0.05	**81.2**	0.31	−0.14	81
0.10	81.0	0.28	**−0.20**	65

5.4 Sensitivity–Persistence Decoupling and Its Mitigation

Protocol. CrossBench4-2000 contains 2,000 questions from Geometry3K, MMK12, LogicVista, and MathVerse, each with four fixed Base responses, and the questions are selected without regard to correctness or span labels. Because teacher-forced scoring of every checkpoint is costly, mechanism results use the model from the first of the three training seeds. Each model scores the same responses and spans under the original image and a hash-fixed masked image. We report direct EFS from Equation 3, where higher values mean greater visual sensitivity, and corrected persistence from Equation 5, where lower values mean stronger retraction. More details are in Appendix C.

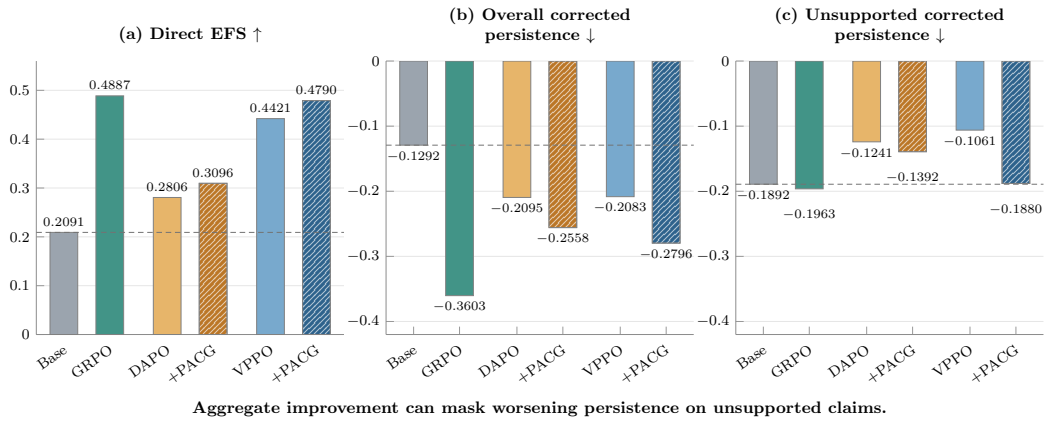


Figure 4: **Aggregate improvement masks unsupported-claim deterioration.** (a) Direct EFS (↑); (b) overall and (c) unsupported corrected persistence (↓). DAPO and VPPO improve aggregate retractability while worsening unsupported persistence; PACG mitigates this deterioration.

Q6: Does higher visual sensitivity come with better retractability? No, and this is the central finding. As shown in Figure 4, DAPO and VPPO both increase EFS and reduce overall persistence, yet their unsupported persistence rises by 0.0651 and 0.0831. GRPO increases EFS the most without a resolved deterioration, so the two quantities are decoupled rather than tied. DAPO and VPPO also

apply roughly three times the effective positive update mass of GRPO to routed claims, which may contribute to this observed difference and suggests that the effect is optimizer-dependent.

Q7: Does PACG fix the decoupling it diagnoses? Partly for DAPO and substantially for VPPO. Table 5 shows that PACG lowers unsupported persistence by 0.0151 and 0.0819 while raising EFS and accuracy. VPPO+PACG returns to the Base level, whereas DAPO+PACG recovers part of the gap. The phase trajectory in Figure 5 shows the same divergence over training. DAPO drifts toward greater sensitivity but weaker retractability, while PACG gains sensitivity without this drift. Image-versus-text DID worsens slightly for DAPO+PACG and improves for VPPO+PACG, so this complementary diagnostic depends on the optimizer. Further analyses are provided in Appendix E.

Table 5: **Claim-level mechanism results.** Accuracy (%) is evaluated on the CrossBench4-2000 questions using seed-0 checkpoints. DID is the Base-relative image-versus-text corrected-persistence contrast. † marks an unsupported-persistence change whose paired 95% question-clustered bootstrap CI excludes zero: versus Base for parent optimizers, and versus the corresponding parent for PACG. Full intervals are in Appendix E.2.

Model	Unsup. Pers. ↓	Img-vs-Text DID ↓	Acc. (%) ↑
Base	−0.1892	0.000	38.9
GRPO	**−0.1963**	**−0.167**	59.0
DAPO	−0.1241†	−0.022	62.2
VPPO	−0.1061†	−0.042	63.4
DAPO+PACG	−0.1392†	−0.016	63.0
VPPO+PACG	−0.1880†	−0.066	**64.3**

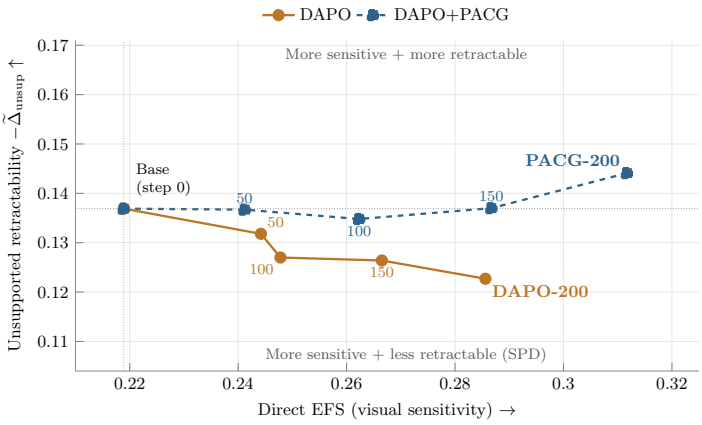


Figure 5: **Sensitivity–retractability phase trajectory.** Direct EFS versus negative unsupported persistence on the fixed 500-question subset. Right: more sensitive, up: more retractable. Lines connect steps $0, 50, 100, 150, 200$; dotted lines mark Base.

6 Conclusion

In this paper, we identify Sensitivity–Persistence Decoupling as an overlooked failure of credit assignment in multimodal RL. Using a fixed-rollout counterfactual diagnostic, we show that DAPO and VPPO make visual claims more sensitive to the image while simultaneously making unsupported claims more persistent. To address this, we propose Persistence-Aware Credit Gating (PACG), which tests whether each direct visual claim retracts when its evidence is disrupted and selectively attenuates the positive credit of claims that do not. Without supported/unsupported labels or inference-time cost, PACG improves unsupported-claim retractability, raises accuracy across nine multimodal reasoning benchmarks for both outcome-driven and perception-aware optimizers, and transfers to a larger scale, a newer backbone, and HallusionBench. Our work shows that whether a policy looks at the image and whether it would take back what the image does not support are fundamentally distinct questions. We believe that holding visual claims accountable to visual evidence, rather than only making them dependent on it, offers a promising direction for multimodal credit assignment.

AI USE STATEMENT

Large language models, such as ChatGPT, were used to assist with drafting, editing, and polishing the manuscript. They were not used for research ideation. In addition, LLM-based coding assistants, such as Copilot, were employed to support code implementation. The authors take responsibility for the final manuscript, code, and reported findings.

ETHICS STATEMENT

This work studies the reliability of visual claims in multimodal reasoning. Although improved claim retractability may help identify unreliable assertions, PACG is not a factuality verifier and does not eliminate hallucinations. Automated annotations can contain errors and biases; our human audit and limitations discussion document these concerns (Appendices D and J). Applications involving consequential decisions require independent validation and appropriate human oversight. Responsible use and release of the associated data and models should respect applicable licenses, privacy, and the ICLR Code of Ethics.

REPRODUCIBILITY STATEMENT

Anonymous code is available at `https://anonymous.4open.science/r/PACGreviewcodeD9A7`. It provides a selective reference implementation of our core ideas, including visual-claim routing and alignment, PACG credit gating, EFS and corrected-persistence computation, and accuracy aggregation, together with synthetic tests and training-integration pseudocode. It is not a complete end-to-end training repository. Training configurations, the full algorithm, fixed-reference construction, and annotation prompts are documented in Appendices A, B, C, and I, respectively. We plan to publicly release the complete code and additional implementation and experimental details following the anonymous review process.

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A Training and Implementation Details

A.1 Shared Training Recipe

We follow the VPPO training setup for Qwen2.5-VL, using ViRL39K, two training epochs, and eight sampled responses per prompt. Qwen2.5-VL-32B uses the same settings as the 7B model, including the PACG parameters, except that its maximum response length is 4,096 instead of 2,048. The Qwen3-VL-2B-Thinking runs use a 200-step budget and a maximum response length of 8,192, with the same entropy penalty of 0.1. The vision tower remains trainable. Rewards are binary answer correctness, without an additional format reward. Unlike the VPPO reference setting's entropy coefficient of 0.06, our reported recipe uses an entropy penalty of 0.1: the minimized loss includes $+0.1\,\mathcal{H}$, rather than an entropy bonus. Table 6 summarizes the settings for all three backbones. Qwen3-VL-2B-Thinking is the additional backbone evaluated in the main results; its results are reported separately from Qwen2.5-VL-7B and Qwen2.5-VL-32B.

Table 6: **Training configurations.** All three backbones use an entropy penalty of 0.1; training budgets and response-length limits are backbone-specific.

Setting	**Qwen2.5-VL-7B**	**Qwen2.5-VL-32B**	**Qwen3-VL-2B Thinking**
Training corpus	ViRL39K	ViRL39K	ViRL39K
Training budget	2 epochs	2 epochs	200 steps
Optimizer	AdamW	AdamW	AdamW
Learning rate	10^{-6}	10^{-6}	10^{-6}
Adam betas	$(0.9, 0.999)$	$(0.9, 0.999)$	$(0.9, 0.999)$
Weight decay	0.01	0.01	0.01
Learning-rate schedule	Constant, no warmup	Constant, no warmup	Constant, no warmup
Freeze vision tower	No	No	No
Rollout batch size	384	384	384
Global update batch size	128	128	128
Responses per prompt	8	8	8
Rollout temperature / top-p	1.0 / 0.99	1.0 / 0.99	1.0 / 0.99
Maximum prompt length	4,096	4,096	4,096
Maximum response length	2,048	4,096	8,192
Entropy penalty coefficient	0.1	0.1	0.1
Training reward	Binary accuracy	Binary accuracy	Binary accuracy
Parameter precision	BF16	BF16	BF16
Gradient checkpointing	Enabled	Enabled	Enabled

A.2 Parent Optimizers and Regularization

DAPO and its PACG composition use dynamic sampling, asymmetric clipping with lower and upper thresholds 0.20 and 0.28, and token-mean policy loss. Plain GRPO does not inherit DAPO's dynamic filtering. PACG preserves the chosen parent's reward and loss configuration and changes only the routed positive token credits. Frozen-reference KL regularization is separate from the entropy penalty and from VPPO's visual-dependency score. Run configurations specify either no reference KL or `low_var_kl` with coefficient 0.01; this setting is separate from the shared recipe in Table 6.

A.3 PACG Configuration

We use retraction margin $m = 0.05$, attenuation strength $\alpha = 10$, and minimum gate $g_{\min} = 0.30$, with $g_s = \max\{g_{\min}, \exp[-\alpha \max(0, \widetilde{\Delta}(s)+m)]\}$. Gating is restricted to routed direct visual claims and positive advantages; negative advantages and non-routed tokens are unchanged. Overlapping spans use the minimum gate. Residual calibration requires at least eight residual response tokens; invalid routing, alignment, intervention, or residual control falls back to unit gates. The training counterfactual is random patch blackening with 14×14-pixel patches and blackening probability 0.5. Supported/unsupported labels are not used to assign gates.

A.4 Composition with VPPO

VPPO retains its visual-token selection fraction of 0.4 and minimum trajectory-level advantage scaling of 0.9, with the upper scaling factor determined dynamically by batch normalization. The visual-dependency score uses the low-variance KL estimator under image perturbation. This score is not a KL regularizer toward a frozen reference model. PACG is applied to the resulting base credits with the same positive-only rule as for DAPO.

A.5 Systems and Evaluation

Training uses the repository's VERL/EasyR1-derived pipeline with FSDP, BF16 parameters, gradient checkpointing, and vLLM rollout generation. Micro-batching and rollout tensor parallelism are deployment settings, not gate hyperparameters; the shared 7B launcher defaults to experience micro-batch size 8 and update micro-batch size 4 per device. The measured 7B timing configuration uses 16 PPU-810E accelerators (Appendix H). Evaluation uses eight stochastic responses at temperature 1.0 and the same answer extraction and verification pipeline across matched models. Evaluation generation limits are distinct from the training response limits above.

B Full PACG Training Procedure

Algorithm 1 summarizes the training procedure. PACG preserves the parent's rollout generation, reward computation, and policy objective, modifying only eligible positive token credits.

Algorithm 1 Persistence-Aware Credit Gating for Group-Relative RLVR

Require: Prompt $x = (q, I)$; frozen rollout policy $\pi_{\theta_{\text{old}}}$; base optimizer; frozen claim router; degradation operator $\mathcal{C}$

Sample rollout group $\{y_i\}_{i=1}^{G}$ from $\pi_{\theta_{\text{old}}}(\cdot \mid x)$
Compute rewards and base token credits $\{A_{i,t}^{\text{base}}\}$ with the chosen optimizer
Initialize $\widetilde{A}_{i,t} \leftarrow A_{i,t}^{\text{base}}$ and $g_{i,t}^{\text{span}} \leftarrow 1$ for all response tokens
for $i = 1, \ldots, G$ **do**
    Route and token-align direct visual claims in y_i
    Construct residual response-token set $\mathcal{R}_i$
    Construct and validate $I_i^{\text{cf}} = \mathcal{C}(I)$
    if routing, alignment, intervention, or residual control is invalid **then**
        Set all span gates for y_i to one
        continue
    end if
    Teacher-force fixed y_i under I and I_i^{cf} using $\pi_{\theta_{\text{old}}}$
    Compute $\widetilde{\Delta}(s)$ and g_s for each routed claim
    Resolve overlapping claims using $g_{i,t}^{\text{span}} = \min_{s \ni t} g_s$; retain one for tokens outside routed claims
    Form $\widetilde{A}_{i,t}$ with the positive-only rule in Equation 7
end for
Update π_θ with the base clipped objective using $\widetilde{A}_{i,t}$

B.1 Routing and Residual Calibration

The frozen router identifies contiguous spans asserting information read directly from the image, including counts, OCR readings, geometric properties, and spatial relations. Directness identifies a span's evidential role; supported/unsupported labels are reserved for analysis. For rollout i, let $\mathcal{V}_i$ contain tokens annotated as direct or derived visual evidence, or as directly or indirectly visually dependent. The residual control set is

$$\mathcal{R}_i = \{t : \ t \text{ is a valid response-text token and } t \notin \mathcal{V}_i\}. \tag{8}$$

PACG requires at least eight residual tokens. Invalid routing, alignment, intervention, or residual control yields unit gates. The annotation instructions and output schema are reproduced in Appendix I.1.

For rollout y_i, a frozen router returns a character interval $s = (a_s, b_s)$, which is aligned to the response-token set

$$\mathcal{T}_s = \left\{ t : [c_t^{\text{start}}, c_t^{\text{end}}) \cap [a_s, b_s) \neq \varnothing \right\}, \tag{9}$$

where $[c_t^{\text{start}}, c_t^{\text{end}})$ is token t's decoded character interval.

The residual set is a practical within-trajectory reference, not a set assumed to be strictly vision-independent. Its mean shift calibrates trajectory-wide drift rather than estimating a pure non-visual baseline. Routing is an eligibility decision, not an error detector: no supported/unsupported label conditions the gate. Image-level degradation likewise need not remove exactly the evidence relevant to a single claim.

B.2 Formal Analysis

We give two propositions and a remark about PACG's credit transformation. They are conditional, fixed-update properties, not convergence guarantees. Changes in future claim persistence are evaluated empirically in Section 5.4.

Write $g(\delta) = \max\{g_{\min}, \exp(-\alpha[\delta + m]_+)\}$. For $\alpha > 0$ and $0 < g_{\min} < 1$, this function is non-increasing, equals one for $\delta \leq -m$, and is strictly decreasing on $(-m, \delta_{\text{f}})$, where $\delta_{\text{f}} = \log(1/g_{\min})/\alpha - m$. On this interval,

$$g'(\delta) = -\alpha g(\delta) < 0. \tag{10}$$

Outside it the derivative is zero, apart from the boundary points. If $\alpha = 0$ or $g_{\min} = 1$, the gate is identically one.

Conditional label-free credit redistribution. Fix a rollout policy and a sampling rule over valid, positively credited routed claim tokens with a single covering claim. Let $A > 0$ be the parent token credit, X the claim's corrected persistence, and $Z \in \{\text{sup}, \text{unsup}\}$ its evidential status. All expectations and distributions below use this same positive-credit population, not an unconditional collection of claims. Set $C_z^{\text{par}} = \mathbb{E}[A \mid Z = z]$ and $C_z = \mathbb{E}[g(X)A \mid Z = z]$.

Proposition 1 (Relative positive-credit retention)**.** *Suppose both status groups have positive probability, $0 < C_z^{\text{par}} < \infty$, and A and X are independent conditional on Z. Suppose also that $F_{\text{unsup}}(x) \leq F_{\text{sup}}(x)$ for all x, where F_z is the conditional CDF of X given $Z = z$. Then*

$$\frac{C_{\text{unsup}}}{C_{\text{sup}}} \leq \frac{C_{\text{unsup}}^{\text{par}}}{C_{\text{sup}}^{\text{par}}}. \tag{11}$$

For $\alpha > 0$ and $g_{\min} < 1$, the inequality is strict if $F_{\text{sup}} - F_{\text{unsup}} > 0$ on a set of positive Lebesgue measure inside $(-m, \delta_{\text{f}})$.

Proof. Conditional independence gives $C_z = C_z^{\text{par}} \mathbb{E}[g(X) \mid Z = z]$. The gate is non-increasing, so the assumed stochastic dominance implies $\mathbb{E}[g(X) \mid Z = \text{unsup}] \leq \mathbb{E}[g(X) \mid Z = \text{sup}]$. For $\alpha > 0$ and $g_{\min} < 1$, their difference in the opposite order is

$$\int_{-m}^{\delta_{\text{f}}} \alpha e^{-\alpha(x+m)} \big(F_{\text{sup}}(x) - F_{\text{unsup}}(x)\big)\, dx \geq 0.$$

The stated strictness condition makes this integral positive. Since every gate is at least $g_{\min} > 0$, division yields the result. □

No support label enters the gate. The conclusion concerns a relative credit ratio, not an absolute ordering of credit between groups. For overlapping claims the actual gate is the minimum of their gates; the proposition applies if its assumptions are instead imposed on the effective persistence $X = \max_{s \ni t} \widetilde{\Delta}(s)$ and an explicitly defined token-status population. Dominance of individual claim persistences alone does not establish dominance of this maximum. Invalid-scoring fallbacks are excluded from Proposition 1.

The assumptions concern the evolving rollout policy and are not established by our fixed-Base-response evaluation. In that evaluation, unsupported persistence is lower than supported persistence

under Base (-0.1892 versus -0.137), whereas the mean ordering reverses under DAPO (-0.1241 versus -0.159) and VPPO (-0.1061 versus -0.184; Tables 5 and 11). The latter ordering is compatible with the proposed condition but proves neither stochastic dominance nor conditional independence, and does not verify either assumption on online positive-credit tokens. The proposition therefore describes a possible operating regime rather than an experimentally established explanation of PACG's gains.

Bounded deviation at a fixed actor update.

Proposition 2 (Credit bounds and unchanged surrogate branches)**.** *Hold the sampled tokens, loss masks, parent advantages, and gates fixed, with no differentiation through advantages or gates. PACG satisfies*

$$\operatorname{sign}(\widetilde{A}_{i,t}) = \operatorname{sign}(A^{\text{base}}_{i,t}), \qquad |\widetilde{A}_{i,t} - A^{\text{base}}_{i,t}| \le (1 - g_{\min})[A^{\text{base}}_{i,t}]_+. \tag{12}$$

At the same policy parameters, PACG preserves the active branches of the parent's clipped surrogate. Wherever its per-token gradient exists, a gated token's surrogate gradient is multiplied by $g^{\text{span}}_{i,t}$. For the token-mean objective in Equation 1, with positive policy ratios and $0 \le \epsilon_{\text{low}} < 1$, $\epsilon_{\text{high}} \ge 0$,

$$|\mathcal{J}_{\text{PACG}} - \mathcal{J}_{\text{par}}| \le (1 - g_{\min})(1 + \epsilon_{\text{high}})\mathbb{E}\left[\frac{\sum_i \sum_{t \in \mathcal{T}_i^+} A^{\text{base}}_{i,t}}{\sum_i |y_i|}\right], \tag{13}$$

where $\mathcal{T}_i^+$ contains routed tokens with positive parent credit. The expectations are assumed finite.

Proof. The credit bounds follow from $g^{\text{span}}_{i,t} \in [g_{\min}, 1]$; all other credits are unchanged. For $A > 0$, write the surrogate as $S(\rho, A) = A \min\{\rho, 1+\epsilon_{\text{high}}\}$. Thus $S(\rho, gA) = gS(\rho, A)$ for a fixed positive gate: branch selection is unchanged and differentiation scales the token gradient by g. At branch ties, the corresponding one-sided derivatives scale likewise. Furthermore,

$$0 \le S(\rho, A) - S(\rho, gA) = (1 - g)A \min\{\rho, 1 + \epsilon_{\text{high}}\} \le (1 - g_{\min})A(1 + \epsilon_{\text{high}}).$$

Summing with the objective's nonnegative normalization proves the bound. □

The same argument applies to sequence-mean GRPO with its corresponding weights in the bound. If tokens are masked, the same fixed masks and normalization must be used on both sides. Unchanged regularizers cancel in an objective-value comparison, but their gradients are not multiplied by the gate. Per-token scaling need not preserve the direction of the summed gradient or the eventual optimizer step. These results do not assert a hard trust region, unchanged future clipping rates, or training stability. Unit gates recover the parent objective, including when no claim is routed or validity checks fail.

Invariance of residual calibration.

Remark 1 (Common additive shifts cancel exactly). For fixed claim and residual token sets with nonempty residual control, suppose $d_{i,t} = \eta_{i,t} + b_i$, where b_i is shared by all response tokens in rollout i. Then Equation 5 gives

$$\widetilde{\Delta}(s) = \frac{1}{|\mathcal{T}_s|} \sum_{t \in \mathcal{T}_s} \eta_{i,t} - \frac{1}{|\mathcal{R}_i|} \sum_{u \in \mathcal{R}_i} \eta_{i,u},$$

which is exactly independent of b_i, whereas the raw shift retains b_i. This cancellation motivates residual calibration and the w/o Residual Corr. ablation (Table 3); it does not remove heterogeneous token-specific corruption effects or certify claim-localized evidence removal.

C Fixed-Reference Evaluation Protocol

C.1 CrossBench4 Reference Construction

CrossBench4 comprises Geometry3K, MMK12, LogicVista, and MathVerse. The primary mechanism evaluation uses 2,000 questions with four fixed Qwen2.5-VL-7B Base responses each, yielding 8,000 trajectories. Question selection is frozen before checkpoint evaluation and is blind to answer

correctness, span presence, evidence labels, and mechanism scores. All four responses are retained, including incorrect answers and responses without eligible direct claims.

The reference builder joins the frozen question list with the Base annotation bank by dataset and sample ID. It requires four successfully annotated, nonempty responses and a resolvable image per question; incomplete inputs raise an error rather than replacing questions.

C.2 Scoring and Aggregation

Mechanism results use the seed-0 checkpoint for each trained model; temporal comparisons use checkpoints from the seed-0 run. Question-cluster bootstrap intervals quantify uncertainty over evaluation questions, not variation across training seeds.

Each checkpoint teacher-forces the same responses and exact-aligned spans under original and hash-fixed intervened images. Rollouts are averaged within question, and paired bootstrap intervals resample question clusters identified by dataset and sample ID. Responses without eligible spans contribute no claim-level observation. Thus, 8,000 is the reference-bank size, not the valid-observation count for every subgroup metric.

C.3 Temporal Subset and Evaluation Scope

Figure 6 and Figure 5 use 500 questions from this bank, retaining all four responses (2,000 trajectories). The selected IDs and source checksum are frozen across checkpoints.

Table 5, Figure 4, and the ablation mechanism metrics in Table 16 use the full 2,000-question reference. The nine-benchmark accuracy results, including Table 3, use free generation instead.

C.4 Benchmark Coverage

Task-level accuracy is evaluated on MMK12, MathVerse, DynaMath, MathVision, Geometry3K, We-Math, LogicVista, MMMU-Pro, and Clever-Count. These benchmarks cover visual mathematics, geometry, counting, and broader multimodal reasoning. Matched models share answer extraction and verification, with eight sampled responses per question in the reported mean-accuracy evaluations. Direct EFS and persistence are evaluated on CrossBench4, rather than on all nine accuracy benchmarks.

HallusionBench evaluation. We additionally evaluate the six Qwen2.5-VL-7B variants on HallusionBench (Guan et al., 2024), using one checkpoint per model rather than averaging over training seeds. Kimi-K3 (Team et al., 2026) judges answer correctness at a judge decoding temperature of 0.6, with one judgment per response. We report the mean of eight evaluations, computing each metric within an evaluation before averaging across the eight repeats. aAcc measures per-answer accuracy; fAcc requires all questions associated with an image to be correct; qAcc requires a question to be answered correctly across its visual variants. These results use LLM judging, unlike the rule-based answer verification in the nine-benchmark suite, and do not enter its reported average.

The default mechanism intervention uses hash-fixed random-grid blackening with 14×14-pixel patches and blackening probability 0.5. Additional corruptions are compared in Appendix E.5.

D Claim Routing and Annotation Audits

D.1 Human Audit

Two annotators label a stratified sample of 500 questions from CrossBench4-2000. We evaluate direct visual-claim identification, direct-versus-derived classification, and supported-versus-unsupported classification. Table 7 separates agreement with human consensus from agreement between the annotators.

Agreement is highest for direct-claim identification and lower for evidence-status classification. PACG uses the former for routing and reserves supported/unsupported labels for offline analysis.

Table 7: **Router and human audit on 500 CrossBench4 questions.** Precision, recall, F1, and the first three κ values use human consensus as the reference.

Task	Precision	Recall	F1	κ
Direct visual claim	0.90	0.87	0.88	0.81
Direct vs. derived	0.85	0.83	0.84	0.76
Supported vs. unsupported	0.82	0.78	0.80	0.71
Human–human Cohen's κ (same task order)				
Direct visual claim	—	—	—	0.84
Direct vs. derived	—	—	—	0.79
Supported vs. unsupported	—	—	—	0.74

D.2 Router Selection

On a separate 200-question expert-annotated set, Qwen3.8-27B gives the highest F1 (0.86). Qwen3.6-35B-A3B reaches 0.84 F1 with 165 versus 425 seconds per batch, a 2.6× latency advantage; we use it as the production router. Recall measures matched expert spans, whereas coverage measures questions with at least one matched expert span.

Table 8: **Span-extractor comparison on 200 expert-annotated questions.** Span coverage is the fraction of questions with at least one matched expert span. Timing is wall-clock seconds per batch on eight A100 GPUs; shading identifies the selected router.

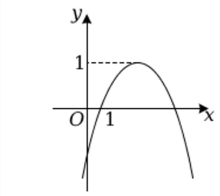
Router	Precision	Recall	F1	Span Coverage	Timing (s)
Qwen3-4B	0.68	0.63	0.65	76%	108
Qwen3.8-27B	0.87	0.85	0.86	94%	425
Qwen3.6-35B-A3B	**0.85**	**0.83**	**0.84**	**92%**	**165**

D.3 Qualitative Examples of Direct Visual Claim Routing

The three examples below use real pre-RL responses from the MMK12 reference bank. Each separates a claim's evidential role from its factual support: both highlighted observations are routed, although only one agrees with the image. The response still reaches the correct final option.

Example 1: A correct answer with a contradicted visual claim

Question (abridged). Given the graph of $y = ax^2 + bx + c$ ($a \neq 0$), which conclusion is correct? Option D states that $ax^2 + bx + c = -1$ has two distinct real roots.



Original response excerpts:

The parabola opens downwards, which means the coefficient a is negative. So, $a < 0$.

[...]

The y-intercept of the parabola is at $y = c$, which is positive

So, $c > 0$.

[...] Given the above analysis, the correct conclusion is: $\boxed{D}$

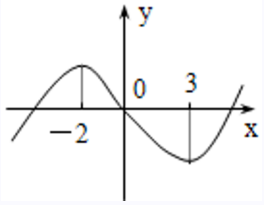
Excerpt	Evidential role	Offline status
Blue observation	Direct visual claim	Supported
Red observation	Direct visual claim	Contradicted
Gray deductions	Derived / non-direct	Not used for routing
Final option D	Answer, not a visual claim	Correct

The graph opens downward and crosses the y-axis below zero, contradicting the red claim. **Red and blue spans are both routed by PACG; their support status is shown only for offline analysis.** The gray text draws conclusions from observations rather than reading new facts from the image.

Ground truth: D. Ellipses mark omissions; punctuation and math typesetting are normalized. Sample: `mmk12_2f53458b788d`.

Example 2: Extrema are not zeros

Question (abridged). From the plotted cubic $f(x) = x^3 + bx^2 + cx + d$, determine where $y = \log_2(x^2 + \frac{2}{3}bx + \frac{c}{3})$ is decreasing.



Original response excerpts:

The function has a local maximum at $x = -2$ and a local minimum at $x = 3$.

The function crosses the x-axis at $x = -2$ and $x = 3$, so -2 and 3 are roots of the polynomial.

[...] $\boxed{A}$

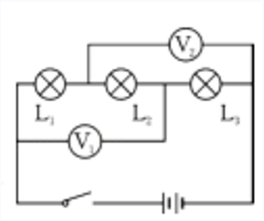
Excerpt	Evidential role	Offline status
Blue observation	Direct visual claim	Supported
Red observation	Direct visual claim	Contradicted
Gray reasoning	Derived / non-direct	Not used for routing
Final option A	Answer, not a visual claim	Correct

The marked locations are turning points above and below the horizontal axis, not crossings of that axis. Both colored claims are routed; support labels remain offline annotations.

Ground truth: A. Ellipses mark omissions; math typesetting is normalized. Sample: `mmk12_e00b8926a68b`.

Example 3: Reading a circuit versus inferring a voltage

Question (abridged). With the switch closed, $V_1 = 7.5\,\mathrm{V}$, $V_2 = 9\,\mathrm{V}$, and supply voltage $12\,\mathrm{V}$, find the voltage across L_2.



Original response excerpts:

The circuit is in a parallel connection.

V_1 measures the voltage across L_1 and L_2.

[...] Therefore, the voltage across L_2 is $4.5\,\mathrm{V}$.

[...] $\boxed{A}$

Excerpt	Evidential role	Offline status
Blue observation	Direct visual claim	Supported
Red observation	Direct visual claim	Contradicted
Gray reasoning	Derived / non-direct	Not used for routing
Final option A	Answer, not a visual claim	Correct

The lamps form a series path. The supported span identifies the two lamps bridged by the voltmeter; the later voltage statement is a computed conclusion. Both colored claims are routed; support labels remain offline annotations.

Ground truth: A. Ellipses mark omissions; math typesetting is normalized. Sample: `mmk12_c10597ae6125`.

D.4 Base Claim Annotation Census

The broader CrossBench4 census contains 5,228 questions and 20,912 Base trajectories, with four responses per question across Geometry3K, MMK12, LogicVista, and MathVerse. It is separate from the selected 2,000-question mechanism reference. The annotation client receives the original images, question, and trace, without the gold answer or source checkpoint identity. Its request format is reproduced in Appendix I.4.

We estimate prevalence from the completed span annotations. Among 8,521 answer-correct trajectories, we count each trajectory once if at least one exact-aligned span has category `hallucinated_or_unsupported_direct_vef`. This yields 2,370 trajectories (27.81%); the category includes both contradicted and unsupported direct visual claims. Exact alignment checks text and character offsets, not label correctness. The confidence threshold of 0.9 applies to a separate high-quality audit subset, not to this prevalence estimate. Exported representative cases are selected for human inspection.

Relationship to the claim-level unsupported rate. The 27.81% reported in the introduction is a *trajectory-level* prevalence: it counts each of 8,521 answer-correct Base trajectories once if *any* of its spans is annotated as `hallucinated_or_unsupported_direct_vef`. The 24.19% in Table 9 is a *claim-level* rate: it is the fraction of qualifying direct claims that are annotated as unsupported, averaged over responses with at least one qualifying claim and then over questions with a defined value. The two quantities also use different populations: the census covers answer-correct Base trajectories across four benchmarks, whereas the free-generation audit covers successfully annotated MMK12 responses, including both correct and incorrect answers, and applies a confidence threshold of 0.9. They are therefore not expected to coincide. The trajectory-level prevalence measures how often a rewarded trajectory contains at least one problematic claim; the claim-level rate measures how often an individual direct visual claim is unsupported.

D.5 Free-Generation Avoidance Audit

We audit each Qwen2.5-VL-7B checkpoint's own free generations on the 2,000-question MMK12 test set, with eight rollouts per question (16,000 expected responses per model). The audit examines whether changes in unsupported visual claims accompany a broad reduction in visual content, shorter answers, or substitution of uncertainty and abstention for claims. All annotation-based metrics, including accuracy, use successfully annotated responses; failed annotations are excluded rather than counted as incorrect. Available rollouts are first macro-averaged within each question. Bracketed intervals in Table 9 are 95% confidence intervals obtained by bootstrapping question clusters.

We annotate these free generations with Qwen3.8-max using the multimodal span-extraction request documented in Appendix I.2 (Prompt 3), supplying the original images, question, and generated response while hiding the source checkpoint's identity.

Direct Claim Rate is the fraction of responses containing at least one qualifying direct visual claim. Direct Claims / Response counts those claims, and Direct Token Ratio measures the fraction of response tokens covered by their union. Claims must satisfy the direct-visual routing criteria, exact text alignment, and annotation confidence of at least 0.9. Unsupported Claim Rate is the fraction of qualifying direct claims annotated as unsupported, averaged over responses with at least one qualifying claim and then over questions with a defined value. Unsupported Response Rate is the fraction of responses containing any such claim. Response length is measured in assistant tokens.

Relative to DAPO, PACG lowers direct-claim rate by 1.60 percentage points and unsupported-claim rate by 2.68 points; mean response length increases from 516.82 to 528.46 tokens. With VPPO, the corresponding rate reductions are 1.51 and 0.83 points, while mean length rises from 681.46 to 697.47 tokens. Median lengths also increase (Table 10). Thus, the lower unsupported-claim rates are accompanied by continued visual-claim generation rather than shorter responses. The reported uncertainty ranges remain descriptive.

Table 9: **Free-generation visual-claim audit on MMK12.** Question-macro means with 95% question-cluster bootstrap intervals, using successfully annotated responses. Rates and accuracy are percentages; lengths are tokens. Uncertainty/abstention entries are point values or ranges, not confidence intervals.

Model	Direct Claim Rate (%)	Direct Claims / Response	Direct Token Ratio (%)	Response Length	Unsup. Claim Rate (%) ↓	Unsup. Resp. Rate (%) ↓	Accuracy (%)	Uncert./ Abst. (%)
Base	77.46 [76.08, 78.81]	2.463 [2.387, 2.537]	11.18 [10.83, 11.54]	506.62 [498.61, 514.84]	24.19 [23.14, 25.27]	32.19 [30.88, 33.48]	43.02 [41.56, 44.51]	10.2
GRPO	78.05 [76.60, 79.48]	2.382 [2.306, 2.459]	11.14 [10.75, 11.52]	526.64 [517.18, 535.97]	19.61 [18.53, 20.73]	26.61 [25.22, 27.99]	73.41 [71.76, 75.07]	9.5–10.5
DAPO	76.12 [74.66, 77.57]	2.361 [2.281, 2.445]	10.30 [9.94, 10.66]	516.82 [508.45, 525.13]	18.84 [17.79, 19.88]	25.83 [24.48, 27.17]	81.60 [79.87, 83.28]	9.5–10.5
DAPO + PACG	74.52 [73.07, 75.96]	2.211 [2.139, 2.282]	9.87 [9.54, 10.21]	528.46 [520.12, 536.96]	16.16 [14.11, 18.25]	25.62 [24.55, 27.10]	83.16 [81.56, 84.75]	9.0–11.0
VPPO	79.41 [78.08, 80.75]	2.645 [2.562, 2.733]	9.95 [9.62, 10.28]	681.46 [670.84, 692.14]	17.44 [16.51, 18.41]	25.61 [24.33, 26.87]	81.73 [80.16, 83.29]	9.5–10.5
VPPO + PACG	77.90 [76.46, 79.28]	2.463 [2.383, 2.543]	9.31 [8.99, 9.62]	697.47 [686.76, 708.45]	16.61 [15.69, 17.55]	24.12 [22.90, 25.37]	82.63 [81.06, 84.14]	9.0–11.0

Table 10: **Response-length medians and annotation coverage.** Coverage is out of 16,000 expected responses per model. Claim counts are exact high-confidence direct claims.

Model	Median length (tokens)	Annotated responses	Direct claims
Base	462.0	16,000/16,000	39,401
GRPO	474.0	15,996/16,000	38,098
DAPO	480.0	15,999/16,000	34,360
DAPO + PACG	489.0	15,998/16,000	35,370
VPPO	635.0	15,990/16,000	42,295
VPPO + PACG	645.0	15,992/16,000	39,375

E Supplementary Mechanism Analysis

E.1 Image-versus-Text Difference-in-Differences

Let $\overline{\widetilde{\Delta}}^{I}_{M}$ and $\overline{\widetilde{\Delta}}^{T}_{M}$ denote the mean corrected persistence of model M on direct visual claims under image and text interventions, respectively. The image intervention holds the question text and response fixed; the text intervention holds the original image and response fixed while perturbing the question text. We report

$$\mathrm{DID}(M) = \left(\overline{\widetilde{\Delta}}^{I}_{M} - \overline{\widetilde{\Delta}}^{I}_{\mathrm{Base}}\right) - \left(\overline{\widetilde{\Delta}}^{T}_{M} - \overline{\widetilde{\Delta}}^{T}_{\mathrm{Base}}\right). \tag{14}$$

Image and text measurements are paired by sample before aggregation. Base therefore has DID zero; comparisons between PACG and its parent subtract their Base-relative DID values. Lower values indicate a more negative image-specific persistence change relative to the text control. The accuracy column in Table 5 is evaluated by free generation on the CrossBench4-2000 questions; it is separate from teacher-forced scoring of the fixed Base responses.

E.2 Supported-Claim Persistence and Bootstrap Intervals

Table 11 complements the main mechanism table with supported-claim corrected persistence on the same fixed reference. Lower values indicate greater relative retractability.

Table 12 reports paired question-clustered intervals for the principal effects. Positive changes in unsupported persistence indicate weaker retractability. For the subgroup interaction $\Delta_{\mathrm{unsup}} - \Delta_{\mathrm{sup}}$, positive values mean greater deterioration on unsupported claims; negative PACG interactions mean greater improvement on that subgroup.

E.3 Online Positive-Credit Exposure

We compare the original GRPO, DAPO, and VPPO parent runs over training steps 181–200. Unlike the fixed-reference mechanism evaluation, this analysis concerns online rollouts used for actor updates.

Table 11: **Supported-claim corrected persistence.** Lower is better. **Bold** and underlining mark the best and second-best point estimates; blue shading identifies PACG variants.

Model	Supported Persistence ↓
Base	−0.137
GRPO	**−0.318**
DAPO	−0.159
VPPO	−0.184
DAPO+PACG	−0.165
VPPO+PACG	<u>−0.213</u>

Table 12: **Paired mechanism effects with question-clustered 95% bootstrap intervals.** Each effect is the first model minus the second. Accuracy effects are on the $[0, 1]$ scale; other effects use their metric's native scale.

Comparison	Metric	Effect	95% CI
DAPO − Base	Unsupported persistence	+0.0651	[0.022, 0.108]
VPPO − Base	Unsupported persistence	+0.0831	[0.036, 0.130]
DAPO+PACG − DAPO	Unsupported persistence	−0.0151	[−0.022, −0.008]
VPPO+PACG − VPPO	Unsupported persistence	−0.0819	[−0.115, −0.049]
DAPO − Base	Direct EFS	+0.0715	[0.025, 0.118]
VPPO − Base	Direct EFS	+0.2330	[0.160, 0.306]
DAPO+PACG − DAPO	Direct EFS	+0.0290	[0.006, 0.052]
VPPO+PACG − VPPO	Direct EFS	+0.0369	[0.010, 0.064]
DAPO − Base	Overall persistence	−0.0803	[−0.122, −0.039]
VPPO − Base	Overall persistence	−0.0791	[−0.121, −0.037]
DAPO+PACG − DAPO	Overall persistence	−0.0463	[−0.078, −0.015]
VPPO+PACG − VPPO	Overall persistence	−0.0713	[−0.110, −0.033]
DAPO − Base	Unsup.–supp. interaction	+0.0872	[0.032, 0.142]
VPPO − Base	Unsup.–supp. interaction	+0.1300	[0.065, 0.195]
DAPO+PACG − DAPO	Unsup.–supp. interaction	−0.0096	[−0.017, −0.002]
VPPO+PACG − VPPO	Unsup.–supp. interaction	−0.0524	[−0.086, −0.019]
DAPO+PACG − DAPO	Image-vs-text DID	+0.0059	[0.001, 0.011]
VPPO+PACG − VPPO	Image-vs-text DID	−0.0246	[−0.045, −0.004]
GRPO − Base	Unsupported persistence	−0.0071	[−0.025, 0.011]
GRPO − Base	Direct EFS	+0.2796	[0.200, 0.359]
DAPO+PACG − DAPO	Accuracy	+0.0079	[0.001, 0.015]
VPPO+PACG − VPPO	Accuracy	+0.0082	[0.001, 0.016]

The frozen PACG router identifies direct visual claims, which are aligned to response tokens. The population includes all routed direct claims, without supported/unsupported filtering.

We report two complementary metrics. The positive routed-token rate, P_{pos}, measures the fraction of routed direct-claim tokens receiving positive parent credit. Effective positive update mass, M_{eff}, measures the positive update mass applied to those tokens. Parent credit is measured before PACG and includes VPPO's own credit shaping. Metrics are aggregated from tokens to rollouts, then to steps, with equal weighting across the 20 steps.

Table 13: **Online positive-credit exposure over steps 181–200.** Results concern routed direct visual claims in the parent training runs. Values are point estimates.

Metric	GRPO	DAPO	VPPO
Positive routed-token rate, P_{pos} (%)	71.8	66.3	78.9
Effective positive update mass, M_{eff}	0.35	0.93	1.00

Table 13 separates the frequency of positive credit from its effective magnitude. Positive routed-token rates are broadly comparable across the three runs, but DAPO and VPPO exhibit substantially greater effective positive update mass than GRPO. In particular, DAPO has a lower positive-token rate than GRPO yet a higher update mass. This pattern is consistent with optimizer-dependent exposure to positive updates, rather than the presence of credit broadcasting alone, being relevant to the observed persistence differences. The comparison is descriptive and does not establish that positive-credit exposure causes SPD.

E.4 Checkpoint Training Dynamics and Sensitivity–Retractability Phase Trajectory

Training dynamics reproduce the endpoint pattern. Figure 6 tracks DAPO and DAPO+PACG from step 0 to 200 on the fixed 500-question subset of CrossBench4-2000. Under DAPO, EFS increases from 0.2189 to 0.2856 while unsupported persistence moves from -0.1369 to -0.1227, showing weaker retraction despite greater visual sensitivity. PACG follows a different trajectory: EFS reaches 0.3117 while unsupported persistence reaches -0.1441. The checkpoint dynamics therefore mirror the endpoint result and show that PACG prevents the unsupported-persistence drift observed during DAPO training.

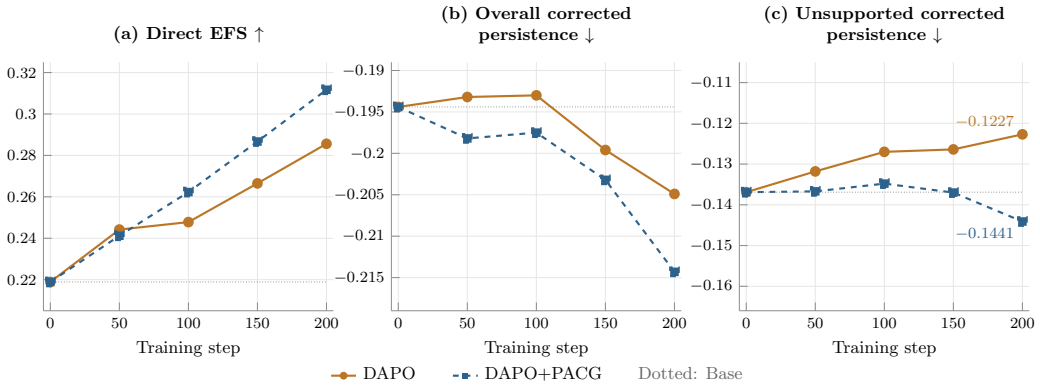


Figure 6: **Sensitivity and unsupported-claim persistence diverge during training.** DAPO and DAPO+PACG are evaluated at matched steps $0, 50, 100, 150$, and 200. Direct EFS increases for both methods, whereas DAPO's unsupported persistence becomes progressively less negative. PACG prevents the same drift and ends with greater unsupported retractability than DAPO. The temporal series uses 500 fixed questions sampled from CrossBench4-2000, with four Base rollouts each; subset averages can differ from full-set averages. Lines connect observed checkpoints rather than fitted curves.

Sensitivity–retractability phase trajectory. Figure 5 combines the direct EFS and unsupported persistence measurements from Figure 6. DAPO ends below and to the right of Base, whereas DAPO+PACG ends above and to the right. Intermediate PACG checkpoints fluctuate around the initial retractability level rather than improving monotonically.

E.5 Robustness to Counterfactual Corruption

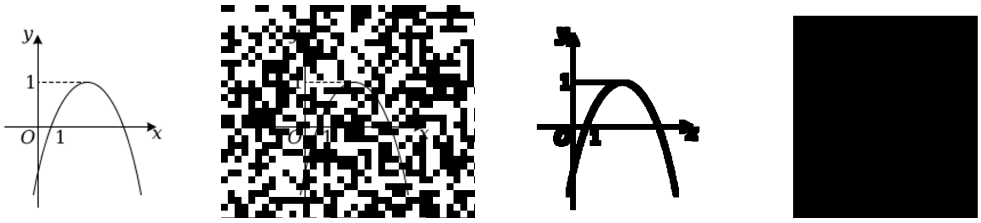
We compare three corruptions using the same fixed reference and checkpoint convention. Table 14 reports absolute PACG measurements and matched changes from DAPO. Valid rate is the fraction of responses for which counterfactual scoring completes.

Table 14: **Robustness to counterfactual corruption.** Absolute DAPO+PACG scores and the matched DAPO+PACG minus DAPO change in unsupported persistence (PACG Effect). Entries are point estimates.

Intervention	EFS ↑	Overall Pers. ↓	Unsup. Pers. ↓	PACG Effect ↓	Valid Rate
Random mask	0.31	−0.165	−0.139	−0.015	96%
Coarse ink	0.28	−0.148	−0.125	−0.013	94%
Foreground/salient	0.33	−0.172	−0.151	−0.018	93%

PACG lowers unsupported persistence under all three corruptions, with effects from -0.013 to -0.018. This consistency supports robustness of the PACG improvement to the tested corruption operators.

How the images are intervened on. Random mask partitions the image into 14×14-pixel grid cells and independently blackens each cell with probability 0.5; a sample-specific hash fixes the random pattern. Coarse ink thresholds the grayscale image at 245 (on the 0–255 scale), expands the resulting dark-pixel mask with a 9×9 maximum filter, and paints the expanded mask black. Foreground/salient uses the bounding rectangle of pixels below the same threshold, pads it on each side by 5% of the larger image dimension (clipped to image boundaries), and blackens the entire rectangle. Thus, "salient" here refers to a coarse foreground heuristic, not a learned saliency map or a claim-specific evidence region.



(a) Original (b) Random mask (c) Coarse ink (d) Foreground box

Figure 7: **Three image interventions on the same example.** Illustrative outputs of the evaluation operators using their default parameters on the original panel recovered from a saved MMK12 audit image. Random mask removes grid cells; coarse ink expands dark strokes; foreground masking removes their padded bounding rectangle. These are image-level controls, not verified claim-localized evidence removals. The illustration does not add observations to Table 14.

F Additional Task-Level Results

F.1 Compatibility with Plain GRPO

We compose PACG with plain GRPO on the same nine benchmarks as Table 1. The GRPO row reproduces that table.

GRPO+PACG raises the reported average from 56.8 to 57.3. The largest displayed gains occur on MMK12 (+1.3 percentage points), We-Math (+1.0), MMMU-Pro (+0.7), and Clever-Count (+0.7); DynaMath changes from 65.1 to 65.0. These results extend the task-level evaluation beyond DAPO and VPPO. The full nine-benchmark component ablation is reported in Table 3.

Table 15: **PACG composed with plain GRPO.** Accuracy (%); GRPO+PACG uses mean accuracy@8. Bold marks the higher displayed value in each column. Entries are point estimates.

Method	MMK12	Math Verse	Dyna Math	Math Vision	Geometry 3K	We-Math	Logic Vista	MMMU-Pro	Clever-Count	Avg.
GRPO	72.2	67.8	**65.1**	30.1	42.0	67.8	47.1	38.3	81.0	56.8
GRPO+PACG	**73.5**	**68.1**	65.0	**30.2**	**42.3**	**68.8**	**47.3**	**39.0**	**81.7**	**57.3**

F.2 Additional Component Mechanism Metrics

The complete nine-benchmark accuracy breakdown and unsupported persistence are reported in Table 3. The table below adds direct EFS and overall corrected persistence under the same reference protocol. Variant definitions follow Table 3.

Table 16: **Additional component mechanism metrics.** Point estimates; lower persistence indicates stronger retraction. † denotes a degenerate EFS value associated with training collapse, not improved visual grounding.

Variant	Direct EFS ↑	Corr. Pers. ↓	Unsup. Pers. ↓
DAPO	0.2806	-0.2095	-0.1241
w/o Residual Corr.	0.4099	-0.2410	-0.1278
w/o Direct Routing	$3.0366^{\dagger}$	-0.1766	-0.0374
w/o Positive-Only	0.2046	-0.2224	-0.1301
PACG (full)	0.3096	$\mathbf{-0.2558}$	$\mathbf{-0.1392}$

F.3 Variation Across Training Seeds

Table 17 reports the nine-benchmark average across three training seeds for each matched Qwen2.5-VL-7B optimizer. Both improvements exceed the reported seed-level standard deviations and are significant under two-sided Welch's t-tests ($p < 0.05$), providing evidence that the gains in Table 1 extend beyond the observed training-seed variation.

Table 17: **Seed variance** (Qwen2.5-VL-7B; nine-benchmark Avg., %; mean $\pm$ sample standard deviation over three training seeds). p: two-sided Welch's t-test between each PACG composition and its parent ($n = 3$ per group); Δ is in percentage points.

Model	Avg. (%)	Δ vs. parent	p
DAPO	58.1 ± 0.40	–	–
DAPO+PACG	59.9 ± 0.52	$+1.8$	0.010
VPPO	59.8 ± 0.40	–	–
VPPO+PACG	60.9 ± 0.47	$+1.1$	0.038

F.4 Sensitivity to the Persistence Margin

We vary $m \in \{0.00, 0.05, 0.10\}$ while fixing $\alpha = 10$ and $g_{\min} = 0.30$. Among these settings, $m = 0.05$ gives the highest MMK12 accuracy (81.2%), an EFS of 0.31, and retains 81% of positive credit under the reported retention metric.

The zero-margin setting matches ungated DAPO's MMK12 accuracy (81.0%) while retaining 95% of positive credit. At $m = 0.10$, retention falls to 65%, EFS to 0.28, and accuracy to 81.0%. Although persistence improves further, the sensitivity and accuracy reductions favor $m = 0.05$ as the joint operating point in this sweep.

Table 18: **Sensitivity to the persistence margin.** Gate strength $\alpha = 10$ and floor $g_{\min} = 0.30$ are fixed. Accuracy and retention are percentages. Shading marks the selected operating point; bold marks the best accuracy, EFS, and persistence values. Entries are point estimates.

m	α	$g_{\min}$	MMK12 Accuracy (%) ↑	EFS ↑	Corrected Persistence ↓	Unsupported Persistence ↓	Positive-credit retention (%)
0.00	10	0.30	81.0	**0.34**	−0.10	−0.08	95
0.05	10	0.30	**81.2**	0.31	−0.23	−0.14	81
0.10	10	0.30	81.0	0.28	**−0.28**	**−0.20**	65

G Online Training Diagnostics

These statistics are measured on each run's own online trajectories. They complement the fixed-reference mechanism evaluation. Each method is shown for one representative seed; faint curves show recorded values and solid curves trailing 10-step means. Sparse GRPO+PACG logs retain their recorded steps without interpolation.

G.1 Gate Dynamics

Across the three PACG compositions, the mean span gate decreases gradually, around half of positive direct-claim tokens are downweighted, and a smaller fraction reaches the gate floor (Figure 8). The token-multiplier mean, logged as positive-credit retention, is not weighted by advantage magnitude.

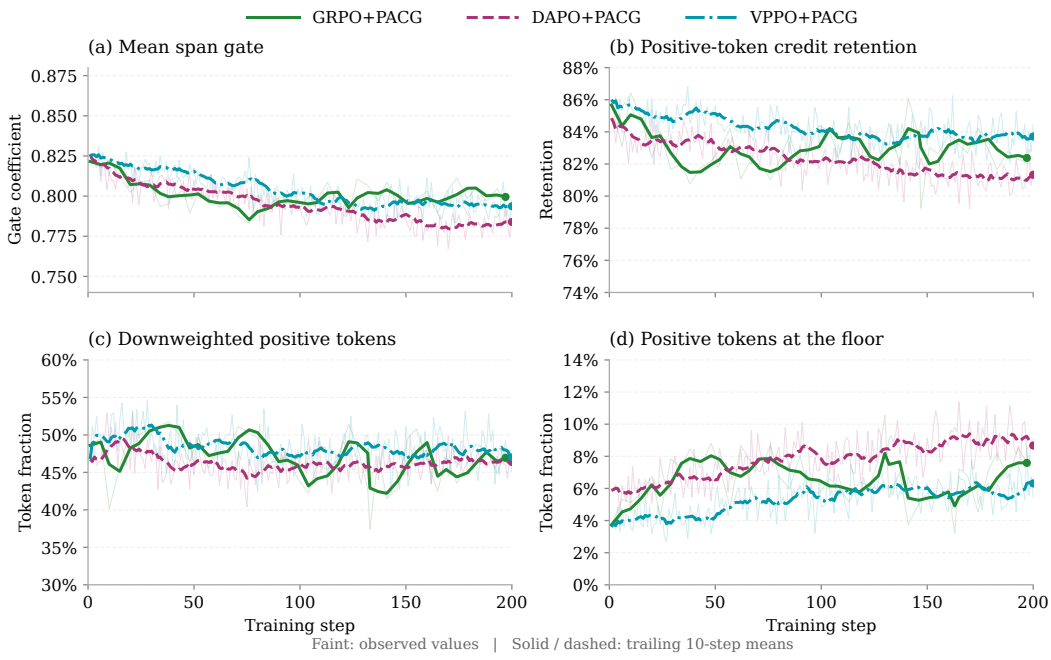


Figure 8: **PACG gate dynamics.** (a) Mean span gate; (b) mean multiplier over positive direct-claim tokens; (c) fraction of these tokens downweighted; (d) fraction at the gate floor. Faint curves show observations and solid curves trailing 10-step means.

G.2 Online Accuracy Reward

Figure 9 compares each parent optimizer with its PACG composition. PACG preserves GRPO's accuracy-reward trajectory and raises the later trajectory for DAPO and VPPO.

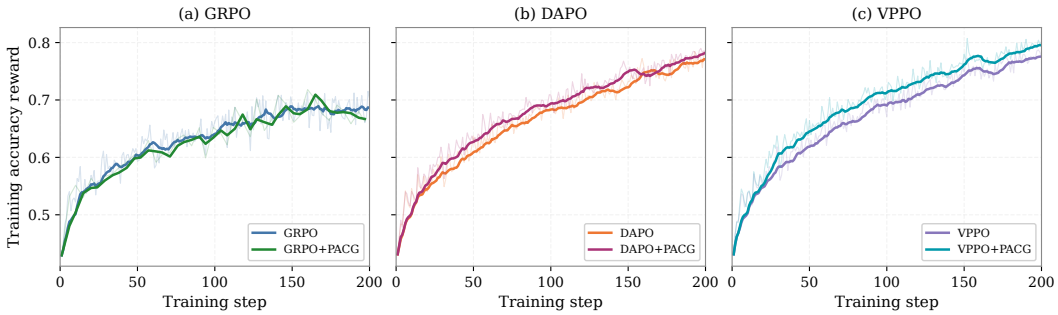


Figure 9: **Online training-reward dynamics.** Paired parent/PACG comparisons of `reward/accuracy` through step 200. Faint curves show logged values and solid curves trailing 10-step means.

G.3 VPPO Visual-Selector Scores and Gate Overlap

VPPO+PACG generally gives higher scores to selected tokens and lower scores to unselected tokens (Figure 10), showing greater separation in the online selector signal. These policy-dependent token populations differ from the fixed direct claims used to measure EFS and retractability.

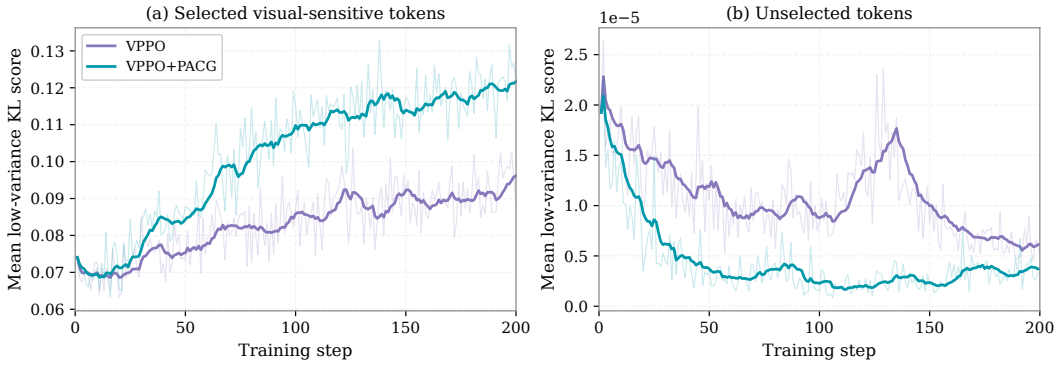


Figure 10: **Online VPPO visual-selector scores.** Mean low-variance KL scores for (a) selected and (b) unselected tokens. The panels use different vertical scales.

The overlap shows that visual-token selection and credit attenuation have intersecting but distinct scopes.

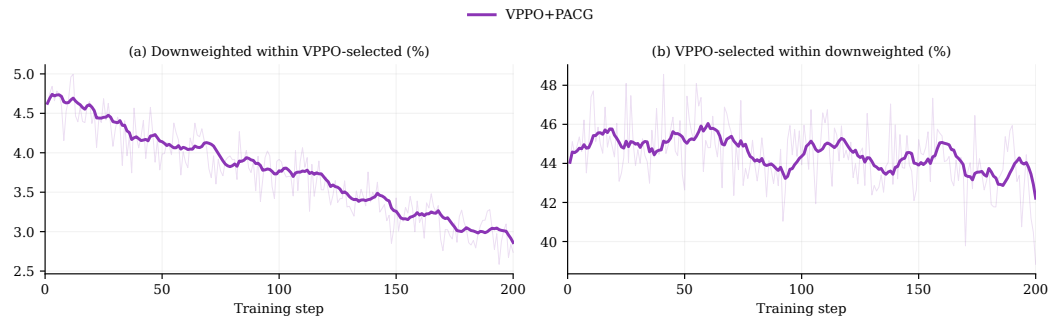


Figure 11: **Overlap between VPPO selection and PACG attenuation.** (a) Fraction of VPPO-selected tokens downweighted by PACG; (b) fraction of PACG-downweighted tokens selected by VPPO. The two panels use different denominators.

G.4 Optimization and Routing Coverage

Figures 12 and 13 report the original exported values through step 200 without offsets or time corrections. DAPO+PACG uses $m = 0.05$. Optimization statistics, routing coverage, and selector overlap provide complementary views of how the gate operates on the evolving response distribution.

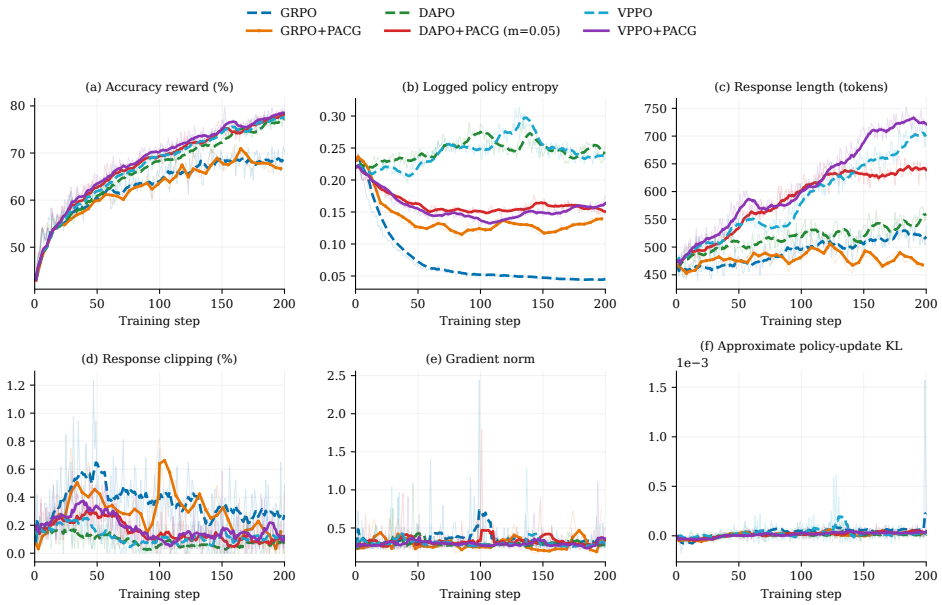


Figure 12: **Training dynamics across six runs.** Accuracy reward, entropy, response length, clipping rate, gradient norm, and approximate policy-update KL. Dashed curves denote parents and solid curves their PACG compositions. Policy-update KL is distinct from reference-model KL and image-intervention EFS.

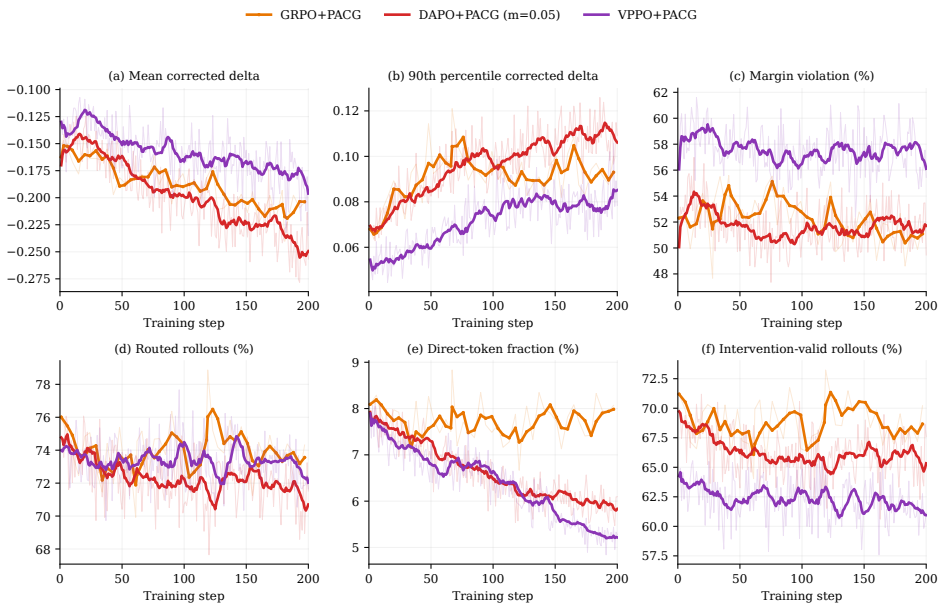


Figure 13: **Online persistence and routing coverage.** (a)–(c) Mean corrected delta, its 90th percentile, and margin-violation rate over eligible spans. (d)–(f) Routed-rollout rate, direct-token fraction, and intervention-valid rate. The 90th percentile describes the span distribution, not a confidence interval.

Figure 13 separates average counterfactual response, its upper tail, and the population entering the gate. Its spans include both supported and unsupported claims.

H TRAINING EFFICIENCY

We compare DAPO and PACG ($m = 0.05$) on Qwen2.5-VL-7B over training steps 1–200 using 16 PPU-810E accelerators. Table 19 gives stage-level and total times.

Table 19: **Training efficiency on Qwen2.5-VL-7B using 16 PPU-810E accelerators.** Mean seconds per step over steps 1–200. Total time includes validation, checkpointing, and other work.

Stage	DAPO	PACG	Overhead vs. DAPO
Rollout	148.8	174.6	–
Original-image teacher forcing	46.7	49.2	–
Reference log-probability (KL)	44.9	–	–
Augmented scoring	–	51.2	–
Counterfactual teacher forcing	–	56.6	–
Blocking annotation wait	–	159.4	–
Actor update	197.1	207.1	–
Total step time (s)	**464.3**	**730.4**	1.57× (+57.3%)
Total logged 200-step time (h)	25.8	40.6	–

Mean step time increases from 464.3 to 730.4 seconds (1.57×, or 57.3% overhead), with 159.4 seconds spent waiting for annotation. The corresponding 200-step totals are 25.8 and 40.6 hours. PACG adds no inference-time computation.

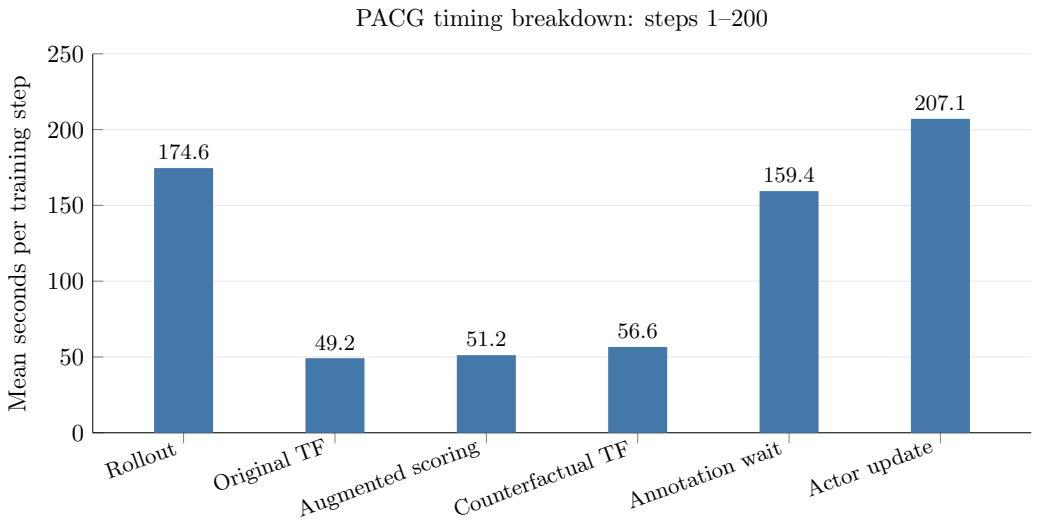


Figure 14: **Training-time breakdown for PACG ($m = 0.05$).** Mean seconds per step over steps 1–200. TF denotes teacher forcing. Augmented scoring corresponds to `timing_s/aug`.

I PROMPTS AND TEMPLATES

The following prompts document span annotation, free-generation auditing, and the standardized reasoning template used in all reported runs.

I.1 MULTIMODAL SPAN ANNOTATION PROMPT

The following boxes reproduce the English span-annotation instructions used in the reported runs. The instruction and output-format portions are separated for readability; they form one text prompt in

the implementation. The request also includes the question's image(s) as image-content items, rather than substituting an image caption. The sample payload contains the sample identifier, question, and trace, truncated to the configured maximum character length. No gold answer is included. These are the exact prompt texts issued in the reported runs.

Prompt 1: Multimodal span annotation instructions

You are a strict multimodal reasoning span annotator. You will see the user question, its corresponding image, and a model-generated reasoning trace.
Task. Identify atomic reasoning spans in the trace and determine which are direct Visual Evidence-Functional (VEF) spans. Do not judge whether the final answer is correct. Do not conflate whether visual evidence actually supports a span with whether that span is direct VEF.

Core principles
- Direct VEF is a functional label, not a correctness label.
- A span can be direct VEF if it claims information read directly from the current image, table, OCR, or spatial layout.
- An incorrect, hallucinated, or image-contradicting visual assertion can still be direct VEF.
- Label the relation to the image separately using the evidence statuses specified below.

A direct VEF span must satisfy all of the following
1. Its text is the smallest complete proposition occurring verbatim and contiguously in the trace.
2. Its function is to read, describe, or restate visual, OCR, chart, counting, attribute, spatial, or topological information in the current image.
3. The proposition cannot be obtained directly from the question text alone; it requires the current image or chart.
4. It is an assertion of direct observation, without calculation, general rules, physical/geometric deduction, option analysis, or a final answer.
5. It is not a hypothetical condition, metadiscourse, or merely a restatement of an answer option.

Not direct VEF
- Facts explicitly stated in the question.
- Hypotheses, such as *If the particle is positively charged*.
- General knowledge, such as *the slope of a velocity-time graph gives acceleration*.
- Mathematical calculations, physical/geometric deductions, derived conclusions, option judgments, or final answers.
- Incomplete phrases, such as *point a to point b* or *from the graph*.
- Pure symbolic reasoning, formula substitution, or answer formatting.
- Sentences describing only the reasoning process or method.
- Meta-judgments about missing, unprovided, or indeterminable information, unless they directly describe visible blank areas, occlusion, or missing regions in the image.

Separate observation from inference. Split *The velocity increases linearly, indicating a constant acceleration* into *The velocity increases linearly* (direct VEF; OCR/chart or visual relation) and *indicating a constant acceleration* (derived VEF or symbolic deduction).

Missing or indeterminable information. Statements such as *we don't have the value of y_5 directly provided*, *cannot be determined*, or *not given* are usually not direct VEF. Only consider direct VEF when the span directly describes a visible blank, occlusion, or missing region; otherwise use derived VEF, non-VEF control, or other.

Evidence status for each direct VEF span
- `supported`: the image clearly supports the proposition.
- `contradicted`: the image clearly negates the proposition.
- `unsupported`: the span claims to read information directly from the image, but sufficient supporting evidence cannot be found, without a clear contradiction.
- `unknown`: image evidence is insufficient to make a judgment.

A contradicted span remains direct VEF. These statuses describe evidential support, not the VEF type.

Analysis categories. Judge visual dependency and consistency with the original image separately:
1. Direct visual dependency, image-supported: `supported_direct_vef`.
2. Direct visual dependency, contradicted or unsupported: `hallucinated_or_unsupported_direct_vef`.
3. No direct visual dependency: `non_vef_reasoning`.
4. Indirect visual dependency, a conclusion derived from a visual observation: `derived_vef`.
5. Direct VEF whose truth cannot be judged from the image: `unknown_direct_vef`.

Do not omit false, fabricated, or image-contradicting visual assertions. If their semantic function claims direct image reading, label them direct VEF and record the lack of support through evidence status and analysis category.

Label vocabulary

`span_group`: `direct_vef`, `derived_vef`, `non_vef_control`, `other`.
`function_type`: `visual_relation`, `spatial_topology`, `ocr_chart`, `counting_attribute`, `uncertainty`, `cross_modal_verification`, `derived_vef`, `symbolic_deduction`, `answer_formatting`, `other`.
`assertion_status`: `asserted`, `hypothetical`, `uncertain`, `quoted_option`.

Annotation rules

- Precision first: when uncertain, do not assign direct VEF.
- Span text must be an exact substring of the trace. Do not paraphrase, translate, or modify LaTeX.
- Keep spans short but complete and independently interpretable.
- Full trace coverage is not required. Return at most 30 spans.
- Do not use the gold answer or consult existing Kimi/Qwen spans.

Checks before outputting any direct VEF span

1. Does it claim to see, read, count, or identify information in the current image?
2. Does it merely repeat the question or answer options?
3. Does it combine what is observed with what is inferred?
4. Does it contain inference cues such as therefore, thus, hence, indicating, implying, so, or which means?
5. If the latter part is removed, does the first part already constitute a complete visual observation?

Prompt 2: Output format and sample payload

Output JSON directly, not Markdown. Use the following JSON format:

```
{
  "sample_id": "...",
  "spans": [
    {
      "text": "exact substring",
      "span_group": "direct_vef",
      "function_type": "ocr_chart",
      "is_vef": true,
      "vef_role": "direct",
      "visual_dependency": "direct",
      "evidence_status": "supported",
      "analysis_category": "supported_direct_vef",
      "claim_polarity": "positive",
      "confidence": 0.95,
      "evidence_in_question_text": false,
      "requires_current_image": true,
      "contains_inference": false,
      "assertion_status": "asserted",
      "rationale": "brief reason"
    }
  ]
}
```

Sample to annotate (placeholders are replaced at request time):

```
{
  "sample_id": "<sample identifier>",
  "question": "<question text>",
  "trace": "<reasoning trace>"
}
```

I.2 Free-Generation Avoidance-Audit Prompt

For the MMK12 free-generation audit in Table 9, we use Qwen3.8-max as the span annotator, following the span-annotation instructions and JSON schema shown in Prompts 1–2. Prompt 3 specifies the message structure used by the annotation client. Each request supplies the original

image(s), question, and generated response, with an opaque identifier that hides the source checkpoint. The client uses temperature 0, top-p 1, and disables thinking. Correct and incorrect responses are both annotated; answer scores are retained separately from the annotation request.

Prompt 3: Free-generation avoidance audit (Qwen3.8-max)

SYSTEM:
You are a strict JSON annotation service. Return exactly one complete JSON object and no reasoning, Markdown, or surrounding prose.

USER (multimodal content):
[Original image(s), supplied as image-content items.]
[The complete span-annotation instructions from Prompt 1, followed by the JSON output schema from Prompt 2.]
Sample to annotate:

```
{
  "sample_id": "<opaque sample identifier>",
  "question": "<question text>",
  "trace": "<freely generated response>"
}
```

Retry suffix (only after an unsuccessful attempt):
Retry `<attempt>`: return exactly one complete JSON object with an explicit `"spans"` list. Every item in spans must be a JSON object. Do not use Markdown.

I.3 Training and Evaluation Prompt Template

For all training and evaluation experiments, we use the single standardized reasoning template below. Its structured format elicits a consistent chain-of-thought response and supports automated parsing of final answers. The question placeholder contains the task question and any answer choices; the corresponding image(s) are supplied as multimodal input.

Prompt 4: Reasoning template

SYSTEM:
You are a helpful assistant.

USER:
`{question}`
You first think through the reasoning process as an internal monologue, enclosed within `<think>` `</think>` tags. Then, provide your final answer enclosed within `\boxed{}`.

I.4 CrossBench4 Base Annotation Request

The CrossBench4 Base annotation pipeline uses the same English instructions and JSON schema shown in Prompts 1–2. It supplies each question's image(s), question text, and generated trace to the annotation model. It does not ask the annotator to grade the final answer or provide the gold answer.

Prompt 5: CrossBench4 Base span-annotation request

SYSTEM (verbatim):
You are a strict JSON annotation service. Return exactly one complete JSON object and no reasoning, Markdown, or surrounding prose.

USER:
`<original image(s), supplied as image-content items>`
`<annotation instructions and JSON schema shown in Prompts 1–2>`
Sample to annotate:

```
{
  "sample_id": "<blind_annotation_id, or request_id if absent>",
  "question": "<question text>",
  "trace": "<trace truncated to max_trace_chars>"
```

```
}
```

Retry suffix (after an unsuccessful attempt):
Retry `<attempt>`: return exactly one complete JSON object with an explicit `"spans"` list. Every item in spans must be a JSON object. Do not use Markdown.

J LIMITATIONS AND BROADER IMPACT

Evaluation scope. The mechanism analysis uses fixed Base responses on four multimodal reasoning benchmarks. This controls the compared claims across checkpoints but does not describe each policy's full generation distribution. The separate nine-benchmark accuracy evaluation and free-generation audit address task performance and generated claim frequency.

Annotation and intervention. Direct-claim routing and evidence-status annotation remain imperfect, as quantified in Appendix D. New domains may require additional annotation calibration. Image corruption measures a claim's response to the chosen intervention, rather than certifying factual correctness or guaranteeing selective removal of that claim's evidence. The robustness comparison covers three operators.

Calibration and compute. The margin sweep supports $m = 0.05$ among the tested settings; it does not jointly tune gate strength and floor across all backbones. Online diagnostics are shown for one representative seed per method. Routing introduces service-dependent training latency. Appendix H reports measured stage-level timing.

Broader impact. Claim-level diagnostics can help identify unreliable visual assertions inside apparently successful reasoning. PACG is a credit-assignment mechanism, not a factuality verifier; deployment still requires task-specific validation, especially in high-stakes domains.

Reproducibility. Training settings, fixed-reference construction, annotation prompts, and additional results are documented in the preceding appendices. Reproducing the mechanism comparisons requires the same frozen questions, responses, spans, and image interventions.

Outlook. Reinforcement learning remains a surprisingly effective and still poorly understood tool for multimodal reasoning. Outcome rewards alone can substantially improve a model's reasoning ability, yet whether the same process teaches the model to *see* better, to ground each statement in what the image actually shows, has been approached from many directions, including perception rewards, visual-dependence reweighting, and process supervision. Our findings add one more piece to this picture: a policy can become more responsive to the image without becoming more willing to withdraw what the image does not support. PACG is one attempt to encourage the latter, so that visual claims are not only visually dependent but also held accountable to visual evidence. We hope that the distinction between sensitivity and retractability, and the phenomena we observe under different optimizers, can inform future work on how multimodal models learn to look before they reason.

K Qualitative Comparisons: Parent Optimizers and PACG

We compare selected responses from DAPO or VPPO with their PACG counterparts on the same question and image. Each pair shows an incorrect parent answer and a correct PACG answer; these selected samples illustrate behavior, not its frequency. The excerpts retain the original wording, with omissions marked by [...]. Red highlights identify erroneous visual assertions; blue highlights identify the relevant contrasting observations. These are manual display annotations, not router outputs. Counting examples concern object inventory rather than the correctness of every attribute description.

The DAPO/PACG pairs precede the two VPPO/PACG pairs. For the cube-folding, histogram, and experimental-setup excerpts, red and green highlights retain the error/correction contrasts of the source qualitative cards.

K.1 Counting small, adjacent airplanes



Question. How many different items are there in the image?

Ground truth: 7 objects.

Generated by DAPO

Incorrect

[...]
1. There is a large gray airplane on the left.
2. There is a small blue and white biplane in the center.
3. There is a large metallic bicycle in the foreground.
4. There is a large gray SUV in the background.
5. There is a large red SUV on the right.
6. There is a small gray bicycle near the red SUV.
[...]
There are 6 distinct items in the image.
$\boxed{6}$

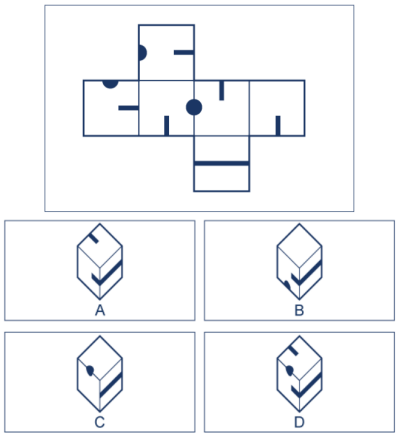
Generated by DAPO+PACG

Correct

[...]
2. List the objects:
- There is a gray airplane on the left.
- There is a shiny silver car in the back.
- There is a small airplane with a teal and cyan body and a red tail.
- There is a smaller airplane with a red and silver body.
- There is a small bicycle with a teal frame and silver components.
- There is a larger bicycle with a teal frame and silver components.
- There is a red car on the right.
[...]
3. Count the total number of unique objects:
- We have one gray airplane.
- We have two cars (one silver and one red).
- **We have two small airplanes.**
- We have two bicycles.
[...]
Therefore, the total number of different items in the image is $\boxed{7}$.

Reading the pair. The image contains three airplanes, two cars, and two bicycles. DAPO lists only one of the two small central airplanes and reports six objects. The PACG inventory includes both small airplanes and reports seven.

K.2 CUBE-FOLDING REASONING



Question. Which cube can be formed by folding the given net?

Choices: A, B, C, D. **Ground truth:** D.

Generated by DAPO

Incorrect

<think>
The three faces represented in option A can be traced to neighboring faces in the net. None appears opposite another, and their line segments seem able to meet round one corner. This **local adjacency check** makes the arrangement appear compatible with the net, so I select option A.
</think>

The predicted answer is $\boxed{A}$.

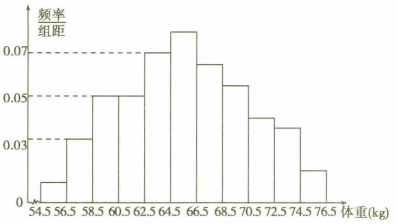
Generated by DAPO+PACG

Correct

<think>
Keep the dotted central face on the left. Its right neighbor folds upward, while the face attached below folds to the right.
The dot remains on the **upper-left edge**, and the two line segments meet at the **front vertical edge**. This preserves the cyclic face order and rejects the mirrored fold.
</think>

The orientation-consistent answer is $\boxed{D}$.

K.3 HISTOGRAM REASONING



Question. Estimate how many of 1,500 boys weigh 56.5–64.5 kg.

Ground truth: 600 boys.

Generated by DAPO

Incorrect

<think>
The interval contains four bars with heights 0.03, 0.05, 0.05, and 0.07. Adding these values gives **0.20**, which is treated as the fraction of boys in the interval.

$$1{,}500 \times 0.20 = 300$$

Using this height-sum interpretation, I would report 300 boys.
</think>

The predicted answer is 300 **boys**.

Generated by DAPO+PACG

Correct

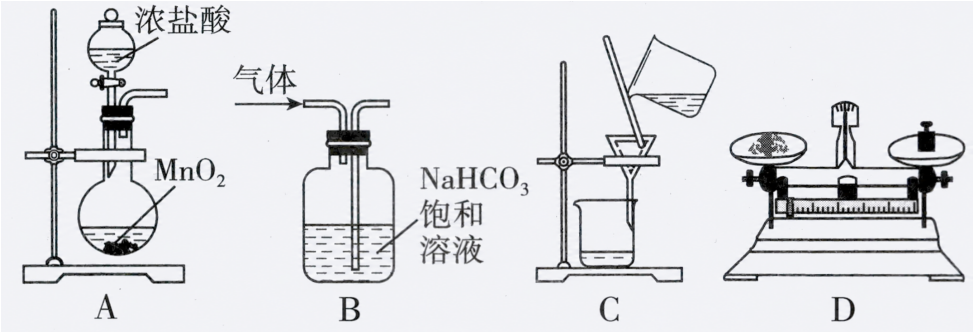
<think>
The vertical axis is **frequency per bin width**, so each bar height is a density rather than a frequency. The selected bins are each **2 kg wide**.

$$2 \times (0.03 + 0.05 + 0.05 + 0.07) = 0.40$$

The total bar area is therefore 0.40, and $1{,}500 \times 0.40 = 600$.
</think>

The evidence-grounded answer is 600 **boys**.

K.4 EXPERIMENTAL-SETUP REASONING



Question. Which apparatus or experimental operation is correctly configured?

Choices: A, B, C, D. **Ground truth:** B.

Generated by DAPO

Incorrect

<think>
In setup A, manganese dioxide reacts with concentrated hydrochloric acid to produce chlorine. The reactants and gas-delivery tube are present, so the apparatus appears suitable for chlorine preparation.
This **reactant-level match** overlooks whether all required operating conditions are shown. I therefore select option A.
</think>

The predicted answer is $\boxed{\text{A}}$.

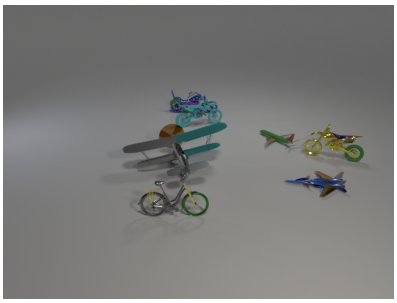
Generated by DAPO+PACG

Correct

<think>
Each setup must be checked against its operating constraints. Setup A is incomplete because the MnO_2–concentrated HCl reaction requires **heating**, which is absent. Setup B correctly passes the gas through **saturated** $NaHCO_3$ **solution** to remove an acidic impurity while retaining the target gas.
The remaining drawings contain procedural inconsistencies, so B is the only configuration that satisfies the full apparatus-level check.
</think>

The constraint-consistent answer is $\boxed{\text{B}}$.

K.5 COUNTING PARTIALLY OVERLAPPING MOTORCYCLES



Question. How many different items are there in the image?

Ground truth: 7 objects.

Generated by VPPO

Incorrect

[. . .]
1. There is a bicycle in the foreground.
2. There is a biplane in the background.
3. There is a small airplane to the right of the biplane.
4. There is a blue motorcycle in the background.
5. There is a blue jet to the right of the small airplane.
6. There is a yellow motorcycle to the right of the small airplane.

[. . .]

Counting these, we have 6 different items.

$\boxed{6}$

Generated by VPPO+PACG

Correct

[. . .]
1. There is a biplane in the foreground.
2. There is a bicycle in the foreground.
3. There is a small green airplane in the background.
4. There is a small blue and purple motorcycle in the background.
5. **There is a small cyan and purple motorcycle in the background.**
6. There is a small yellow and purple motorcycle in the background.
7. There is a small blue and purple airplane in the background.

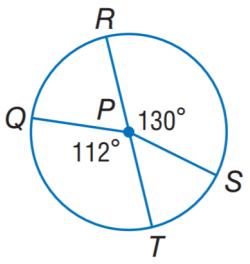
[. . .]

There are 7 different items in the image.

$\boxed{7}$

Reading the pair. The image contains three motorcycles, three airplanes (including the biplane), and one bicycle. VPPO lists only two motorcycles. The PACG response also identifies the cyan motorcycle and reports seven objects.

K.6 READING THE ANGLE ASSOCIATED WITH THE TARGET ARC



Question. In $\odot P$, the radius is 2 inches, find the length of $\widehat{RS}$. Round to the nearest hundredth.

Ground truth: 4.54 inches.

Generated by VPPO

Incorrect

[...]

Notice that the problem provides two central angles: $\angle RPM = 130^\circ$ and $\angle QPM = 112^\circ$.

[...]

The central angle subtending the arc $\widehat{RS}$ is 130°, and the central angle subtending the arc $\widehat{QT}$ is 112°.

[...]

Now we use the formula for the arc length. The radius of the circle is given as 2 inches, and **the central angle corresponding to the arc $\widehat{RS}$ is 118°.**

[...]

The answer is $\boxed{4.12}$.

Generated by VPPO+PACG

Correct

[...]

- The radius r of the circle is 2 inches.
- **The central angle θ for the arc $\widehat{RS}$ is 130°.**

[...]

Here, $\theta = 130^\circ$ and $r = 2$ inches. Plugging in these values, we get:

$$L_{RS} = \frac{130^\circ}{360^\circ} \times 2\pi \times 2$$

[...]

Thus, the length of the arc $\widehat{RS}$ is $\boxed{4.54}$ inches.

Reading the pair. The diagram labels $\angle RPS = 130^\circ$; no point M is shown. VPPO invents angle labels involving M, then uses 118° for the target arc despite initially recognizing 130°. The selected PACG response uses 130° and obtains 4.54 inches.